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Machine and deep learning for personality traits detection: a comprehensive survey and open research challenges

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Abstract
Natural language processing (NLP), a prominent research domain of Artificial Intelligence (AI), analyzes users' generated content on social media for various purposes like sentiment analysis, text summarization, chatbots, fake news detection, etc. Recent advancements in NLP have helped for analysis of human behavior analysis and predicting various human personality traits. Understanding personality traits has long been a fundamental pursuit in psychology and cognitive sciences due to its vast applications for understanding from individuals to social dynamics. Due to online social platforms where people express their views, experiences and comments, NLP is applied for users' behavior and personality analysis, which is helpful in defining marketing strategies, consumers' behavior analysis, team building, etc. This research study provides a comprehensive overview of existing methodologies, applications, and challenges in the field of personality traits detection using shallow machine learning, ensemble learning and deep learning. To conduct this study, recent research publications relevant to NLP for this new but emerging research domain are reviewed. The background knowledge of personality models of various nature is discussed for better domain understanding. The study encompasses machine learning and deep learning models with thorough analysis of traditional and innovative techniques including ensemble learning and transformer-based models in chronological order highlighting the trend analysis showing evolution of application of advanced methods. The review also presents and compares the widely used datasets which may guide the researchers for selection of datasets in future studies. Performance evaluation metrics have been discussed which are used in the relevant literature. Furthermore, it explores the application of research of personality traits detection in various domains highlighting its significance. We have also carried out extensive empirical analysis using conventional textual to advanced deep embedding features and applying machine learning, ensemble learning and deep learning algorithms. Finally, before conclusion, the review highlights the open research issues and challenges as potential future directions for the researchers.

Keywords Artificial intelligence · Machine learning · Personality traits · Data analytics · Psychological analysis · Natural language processing

Extended author information available on the last page of the article

Abbreviations

AI	Artificial intelligence
Bi-LSTM	Bi-directional long short-term memory
BERT	Bidirectional encoder representations from transformers
DT	Decision tree
DL	Deep learning
EEG	Electro encephalo gram
GB	Gradient boost
KNN	K-nearest neighbor
LR	Logistic regression
LSTM	Long short-term memory
MBTI	Myers Briggs type indicators
ML	Machine learning
NLP	Natural language processing
NB	Naïve Bayes
RF	Random forest
SVM	Support vector machine
ViT	Vision transformer
XGB	Extreme gradient boosting

1 Introduction

Understanding human personality is quite a complex and complicated task due to its dynamic nature and it involves an individual's characteristics including patterns of thoughts, emotions, and behaviors. Such attributes also define how people behave and interact with each other as the users in a digital environment using various platforms. Moreover, it also plays a significant role in shaping human interactions, decision-making, and overall psychological well-being (Han et al. 2020). Based on psychology, personality theories have evolved from trait-based models to frameworks that tries to understand and categorize individual differences in thoughts, emotions, and behaviors (Feher and Vernon 2021). Over time, these theories have expanded to include different perspectives extending deeper understanding of how personality shapes behavior, cognition involving reasoning and social engagement (Kuchhal et al. 2022). Personality models with different traits show the cognitive level of individual, the widely used traits are agreeableness, conscientiousness, extroversion, openness, and neuroticism (Ahmad et al. 2020). The study of personality traits is a focus in psychology and leads to the innovation of different frameworks to understand and categorize these traits (Feizi-Derakhshi et al. 2022). Each of these traits represents a spectrum to understanding personality differences. The big-five framework is a descriptive model that many psychologists use to assist in understanding and evaluating personality using Machine Learning (ML) and Deep Learning (DL) detection. The idea of personality refers to an individual's emotions and conduct may both influence how they act (Sikström et al. 2025). Personality model is a theoretical framework, though personality framework data used to describe, categorize, and understand human personality traits and behaviors (Nisha et al. 2022). Personality datasets do not directly measure emotional stability or neurotic tendencies. Therefore, the two systems are not directly equivalent, and the mapping should be seen

as a general precise correlation, that are used as dataset to detect personality traits from online platforms (Ontoum and Chan 2022). The exploration of personality encompasses various perspectives, including biological, cognitive, and learning theories and trait-based approaches. Additionally, it includes the psycho-dynamic and humanistic methodologies in its study. It has had a great deal of interest in current years because of its potential uses in a variety of sectors such as healthcare (Ramachandran et al. 2020), marketing-leadership style (Meskelis and Whittington 2020), political and voting intentions (Zarouali et al. 2024) and human–computer interaction (Sharan and Romano 2020).

The surge in social media usage in recent years has sparked considerable interest in automatic personality recognition from online social media (Hassanein et al. 2018) by predicting user profiles to other group to analyze individual's data such as behaviors detection, sentiment analysis and many more (Hayat et al. 2019), also observe the affects during COVID-19 (Zacher and Rudolph 2021)—(Flesia et al. 2020), by analyzing text, images, and interaction patterns on Facebook, Twitter, and Instagram. Features such as language use, post frequency, and network characteristics are examined to infer human behavior relying on personality traits (Naz et al. 2024a). Different models can manage complex social networks and unstructured data, learning from textual and visual cues to create nuanced personality profiles (Qin et al. 2022). A web-based framework is also designed to predict the different models of personality (Iskandar et al. 2021) with Twitter datasets, developed using the Software Development Life Cycle framework and combines ML techniques with feature extraction methods to analyze tweets for personality type prediction (William et al. 2023). Psycho-linguistic features use language model embedding for personality prediction, leveraging the based on various available personality datasets. It highlights a learning model that outperforms existing methods by blending theory-based text features with automated learning features, offering new insights into language's role in personality assessment (Mehta et al. 2020a).

There is significant interest in developing NLP models capable of autonomously identifying a person's personality traits for the examination of language use, which is a significant demonstration of personality characteristics (Pradhan et al. 2020). Due to a lack of learning from experiences, ML classifications in personality psychology, for handling high-dimensional behavioral data, underscores the potential for more objective and automated personality assessment (Khan et al. 2020). Applying DL to larger data sets markedly boosts the potential of predictive analysis in decision-making (Rahman et al. 2019). Hence, integrating AI computational models helps to implement more compatible and flexible framework to explore personality trait, in various fields. This fusion of AI and psychology brings up new likelihood for perception of human attitude from grounded theory to computational powered models in an online ecosystem (Bousalem et al. 2025).

2 Research motivation

The motivation to determine and study the characteristics of personality arises from the merging of theoretical and practical benefits that are used to understand the complications of human behavior and emotions as well as the study of their cognitive level. This curiosity helps in revealing the facts that help in shaping personality discoveries according to the individual traits and differences involving their personal thoughts and it focuses on the

mechanism that controls human societal interaction. The application of NLP techniques helps in prediction of various traits of personality traits of individuals. This analysis of personality helps in finding proper individuals for marketing purposes, finding customers regarding products and services (Jie et al. 2022), career counselling (Silpa et al. 2024) and other aspects of personalization (Sarwoko and Nurfarida 2021). The dramatic changes in personality research are supported with advancements in computational techniques and an increase in large data. With the inventions and development in ML algorithms, data driven models and predictive analysis, they all have brought changes in the study of personality thus offering unparalleled chances and opportunities in revealing hidden patterns, relationships within a vast number of datasets including social media and digital footprints. In this scenario exploring personality traits limitation are underexplored including LLM, attention models, language-based dataset based on existing studies, comparatively explore in Sect. 4, however detailed analysis of research challenges explored in Sect. 8, ensures that it will enhance our knowledge and understanding of real-world challenges in all fields.

3 Research questions

The following research questions are to be followed to justify the review context for personality trait detection, and their section-wise answers mentioned in Table 1.

RQ1: Which ML Models are more suitable for Personality Traits detection in online textual content?

RQ2: Which DL Models are more appropriate for Personality Traits detection in online textual content?

RQ3: How do diverse features and machine learning based models impact the accuracy of personality trait prediction?

RQ4: What are the features and source of available dataset on personality trait prediction?

RQ5: What are the most potential application areas for AI-based personality trait prediction and how do they impact various fields?

RQ6: What are the main open research challenges and their applicable strategies to be employed to address these issues?

Table 1 Relating section for answers to research questions

Research questions	Answers
RQ1	Section 4; Heading 4.2: Review of Existing Studies using Machine Learning Models
RQ2	Section 4; Heading 4.3: Review of Existing Studies using Deep Learning Algorithms
RQ3	Section 6: Experimental Results
RQ4	Section 7; Datasets
RQ5	Section 8; Applications
RQ6	Section 9; Open Research Issues

4 Research contributions

The implications of personality traits research need a comprehensive study for future researchers, and practitioners get to know more about the intersection of personality psychology and computational methodologies. After evaluating the existing studies, the following review provides the following contributions including:

- Established a novel framework that integrates psychological theories with computational linguistics, enabling the development of more contextually aware and psychologically informed NLP models for personality detection in textual data.
- Discussed a detailed knowledge of personality traits, its model, and relevant background information of data processing along with features and computational models.
- Carried out a brief discussion on identification of methodological trends regarding ML and DL models along with perspective of state-of-the-art models.
- Conducted a comparative analysis of existing studies with our findings based on empirical analysis to define the contribution of this review.
- Provided a comprehensive analysis of key datasets with availability source and learning models also discusses performance measures.
- Examined a brief and quick discussion on potential applications of personality traits in various fields.
- Explored open research challenges can serve as future research directions for the researchers in this new and active domain.

4.1 Comparison with existing relevant review articles

In this literature, comparison presented to show the current literature on personality trait detection has evolved across multiple dimensions, identifying both advancement and gaps spanning from 2020 to 2024 based on numerous studies have explored various ML and NLP to predict traits from textual data, the majority have primarily relied on only ML and DL approaches with minimal list of multimodal data including text, image, audio, and video. However, the potential of NLP review focused on textual data, which capture semantic and contextual information of text, remains a gap in this domain. One significant limitation observed in the existing literature on personality traits prediction is the dearth of literature leveraging computational models with background information including analysis of features and preprocessing techniques to integrate with psychology lens. Moreover, incorporation of psychological models also varies. While some studies, one of them from 2020, lack detailed discussion on personality models, our survey article thoroughly integrates them, reflecting the growing recognition of psychological theories in computational personality detection. Therefore, the absence of research integrating represents a notable gap in current literature, prominence the need for future investigations to discover this advanced text representation approach for more accurate personality traits model prediction, we conduct a systematic literature review by covering all aspect into one article including background detail of psychological personality model as well as computational models. A key observation is the count of discussed text datasets, with earlier studies tending to focus on fewer datasets ranging from 2 to 5, whereas the 2024 survey dataset finds 14 textual datasets with their source link for accessibility, suggesting more extensive datasets exploration. Fur-

thermore, we discuss research challenges, significance of literature with motivations, and applications also covered. All studies address minimal exploration of applications and challenges, in contrast, our 2024 survey provides a comprehensive analysis of complexities in personality detection research. A detailed analysis with the help of Table 2 shows our work and research gap by comparing existing studies.

The paper is following the structured as. Section 2 presents experimental setup and data acquisition. Section 3 discusses personality models. Section 4 investigates the existing studies with machines and deep learning algorithms that are being used for detecting personality traits, also highlights the feature engineering strategies in the context. Section 5 provides analysis of evaluation metrics. Section 6 shares the analysis of experimental results and discussion using various ML and DL models. Section 7 describes the publicly available datasets that exist in the personality domain. Section 8 analyses the most relevant real-world applications. Section 9 highlights the current research issues. Section 10 briefly answers the research questions, Lastly, Sect. 11 concludes our paper. The mind map diagram of review article as shown in Fig. 1.

5 Research plan and search strategy

This systematic review article aims to inclusive examine and categorize the most efficacious techniques, measures, tools, and classifying methods for predicting personality traits. Detailed experimental setup and methodology will be used to synthesize existing accomplishments in the field of personality trait prediction. By observing primary studies, we aim to identify and classify the different range of methodologies and approaches used for predicting traits. The review process will not only merge current understanding but also spot research challenges, providing intuition for future investigations (MacFarlane et al. 2022). To orderly gather relevant data samples for the review on personality trait prediction, a thoroughly organized search strategy is used. This study conducted based on the last five-year literature published in well-reputed top journals, shows the exploration encloses a variety of sources, including academic research papers, case studies, and references from related publications. Additionally, an examination of related websites specializing in personality psychology, psychological evaluation and research institutions is obtained. The methodology extends to incorporating government reports and statistics, akin to the utilization of models. The search process is not only confined to academic literature but also involves an exhaustive review of reference lists from identified publications. The keywords used for the search are keenly selected to ensure a comprehensive extraction of information related to personality behavior prediction. By adopting the systematic approach described in Table 3, the aim is to get meaningful stats from a diverse collection of sources, ensuring the inclusion of worthy perspectives and methodologies in the detailed review of personality trait prediction studies. The search string for personality trait prediction studies can be constructed as follows:

- (Personality ∗ OR MBTI ∗) AND (Prediction ∗ OR Detection ∗)
- (Psychological ∗ OR Psychometric ∗) AND (Personality ∗)
- (Behavioral ∗ OR Psychological ∗) AND (Prediction ∗ OR Assessment ∗)
- (Trait ∗ OR Sentiment ∗) AND (Machine Learning ∗ OR Predictive Modeling ∗)

Table 2 Analysis of proposed literature with existing studies

References	Year	Methodology of existing studies				Computational background discussion				Personality models	Datasets	Applications	RQ	Research challenges
		ML	Ensemble	DL	Transformer	Pre-processing	Features	Models	Evaluation measures					
Ahmad et al. (2020)	2020	✓	x	✓	x	x	X	x	x	✓	4	x	✓	✓
Mehta et al. (2020b)	2020	✓	x	✓	x	x	X	x	x	x	5	✓	x	x
Feher and Vernon (2021)	2021	x	x	x	x	✓	✓	x	x	✓	2	x	✓	x
Feizi-Derakhshi et al. (2022)	2022	✓	x	✓	x	x	✓	x	✓	x	5	x	x	x
William et al. (2023)	2023	x	x	x	x	✓	✓	✓	x	x	2	x	✓	✓
Singh and Singh (2024)	2024	x	x	✓	x	x	X	✓	x	x	2	✓	x	✓
This study	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	14	✓	✓	✓

- (Psychological ∗ OR Personality ∗) AND (Classification*)
- (Trait ∗ OR Behavioral ∗) AND (Text Analysis ∗)

Below search string captures a wide range of studies related to personality trait prediction, including those employing psychological, behavioral, and ML approaches. The use of various terms ensures a comprehensive exploration of literature in the domain.

In our preliminary investigation on databases such as IEEE Explore, Springer, DBLP, ACM, MDPI, and Google Scholar we conducted an exhaustive search for personality trait prediction studies. The initial search yielded a pool of 305 research papers. Through a meticulous screening process based on duplication and titles, we selected 270 papers deemed relevant to study. Subsequently, a thorough analysis of the abstracts led to the identification of an exclusion of 184 research papers. These extracted papers experienced a comprehensive review, resulting in the final of 171 papers as the primary focus of our research. The entire selection procedure delineated in Fig. 2. The comprehensive analysis of the selected research papers involves a rigorous assessment of their quality. Quality control questions, including the research domain and publication in reputable journals or conferences, and keywords selection (Liu et al. 2021), are employed to examine the credibility of the selected studies. Systematic research follows a structured approach in assessing the many methodologies employed for personality behavior prediction. The inclusive categorization is based on prediction algorithms, psychological frameworks, and the combination of machine, deep or hybrid approaches to learning techniques. A tabular summary expresses the characteristics of existing personality trait prediction systems provided. The research strategy guiding our study is systematically addressed through a diligent review of the selected papers. Additionally, the potential and limitations of the studies are critically evaluated and suggestions for future work directions are formulated. The extensive inquiry intends to contribute to the current body of information on personality trait prediction as well as provide fantastic insights for both scholars and practitioners in the field.

6 Relevant background: personality models

Personality models present as frameworks to understand and classify human traits, behaviors, and tendencies, offering incredible insights into individual differences to psychological personality measures (Oravecz et al. 2020). Here, we explore some of the personality views investigated in the relevant literature. The following models are followed in context of psychology encompassing fundamental concepts and methodologies essential for developing and evaluating the framework. Incorporating these models alongside other well-established frameworks as illustrated in Table 4, enriches our understanding of the diverse and multifaceted nature of human personality by highlighting each trait focus characterized by high level and low-level score. Each behavior falls into multiple models as mapped in table, plays a significant role in shaping individual behavior and interactions with others. The high levels often indicate a more pronounced expression of the trait, leading to specific behavior that reflects that intensity, while the low-level score typically represents a more subdued or contrasting approach. This framework provides a concise overview of how these traits influence personality, decision-making, and interpersonal dynamics.

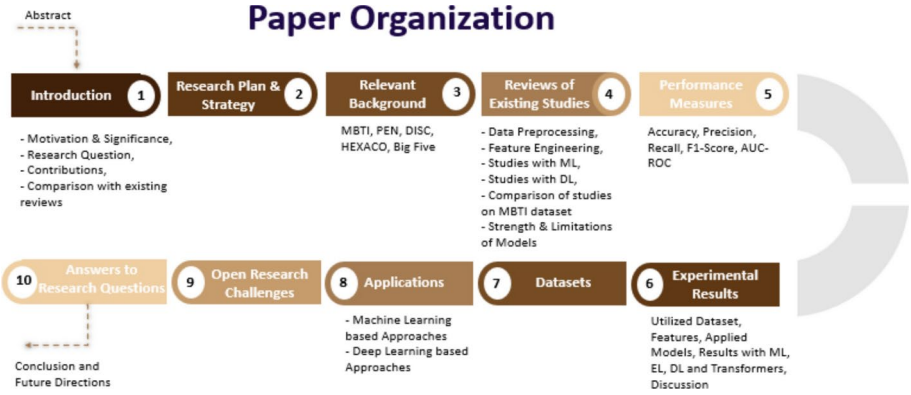

Fig. 1 Structure of survey paper

6.1 MBTI model

One broadly useful model is the Myers-Briggs Type Indicator, which distinguishes sixteen well-defined personality types consisting of four divisions: judging/perceiving, extroversion/introversion, sensing/intuition, and thinking/feeling. Each personality is defined by a unique four-letter code such as INFJ or ESTP, reflecting these preferences. The widely used model in various fields, including psychology, counselling, and organizational behavior to understand and predict the individual differences in behavior, decision-making, and interpersonal interactions that how they perceived the world and preferences (Butt et al. 2025).

6.2 Big five model

Also familiar as the Five-element Model, shows another dominant model, encompassing neuroticism, agreeableness, extroversion, conscientiousness, and openness. Each dimension represents a spectrum, with individuals falling at various points along each continuum. The model gives an inclusive and widely acquired framework for assessment and categorizing personality attributes along these five ways (Mayer and Bryan 2024).

6.3 HEXACO model

The model maintains human personality distilled into six dimensions, admits cultural variations in personality. The model provides a more understanding of personality by incorporating the dimension of honesty and humility, which influences interpersonal and ethical concern. This model map to six personality traits such as Honesty-Humility, Emotionality, Extroversion, Agreeableness, Conscientiousness, Openness. Each of these model traits put up atomically to the concern of human personality, allowing researchers to find out the richness and complexity of individual variance in many contexts (Ashton et al. 2024).

Table 3 Search strategy and formulation

Search	Set of Keywords
Personality*	Personality, personality behavior, personality behavior techniques, personality model
MBTI*	MBTI, MBTI indicators, MBTI classification, MBTI detection
Prediction*	Personality trait prediction, Sentiment prediction
Big five*	Big five personality, big five classification, big five and social media
Detection*	Detect personality traits, personality detection techniques
Psychological*	Psychological parameters, psychological measures, psychological conditions
Psychometric*	Psychometric parameters, psychometric measures, psychometric conditions
Behavioral assessment*	Behavioral prediction, behavioral impact
Sentiment*	Sentiment analysis, sentiment factors, sentiment text prediction
Machine learning*	Machine learning techniques for personality profiles, machine learning methods for detecting traits, Machine learning algorithms for recognizing sentiment prediction
Predictive modeling*	Predictive modeling of personality effects, predictive modeling in psychology, predictive modeling for trait identifications
Classification*	Classification models for psychological traits, classification techniques for trait identification, Classification-based personality assessment, classification frameworks for personality, classification algorithms for personality
Text analysis*	Text and behavioral analysis of personality, text analysis of personality trends, text analysis: personality classification, text analysis of personality insights
Predictive analytics*	Predictive analytics-driven trait detection, predictive analytics in psychological profiling, predictive analytics for personality categorization

6.4 PEN model

Also called the Psychoticism-Extroversion-Neuroticism model, enlarges the Big Five Personality behavior by embodying a dimension of Psychoticism traits such as aggression, impulsiveness, and sensation-seeking. Therefore, in the PEN model, ones are evaluated on three main perspectives: Psychoticism, Extroversion, and Neuroticism. The model gives a subtlety understanding of personality by incorporating traits related to risk-taking, and emotional stability (Netter et al. 2021).

6.5 DISC model

On the contrary, targets observable behavior in various perspectives. DISC represents four main personality traits: Dominance, Influence, Steadiness, and Conscientiousness. Individ-

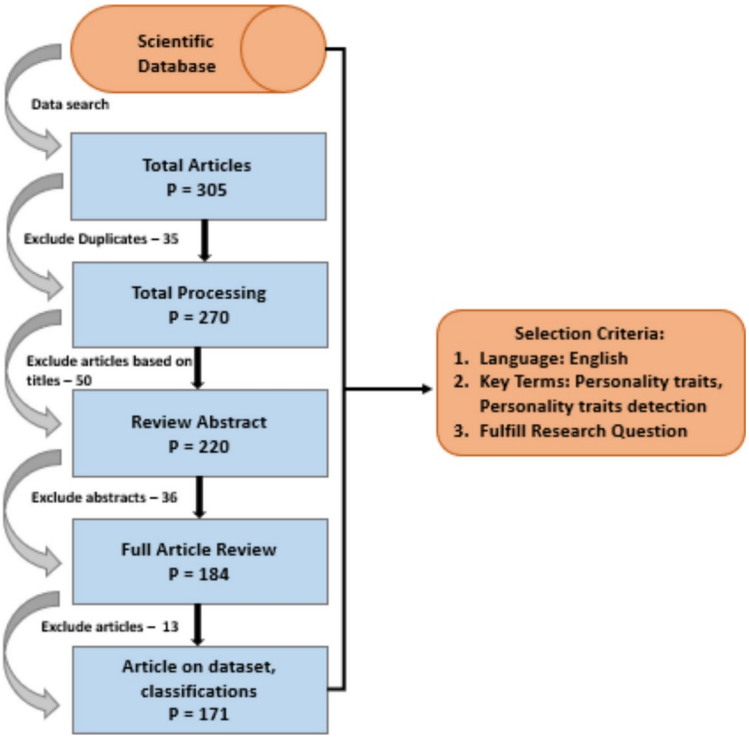

Fig. 2 Flow of dataset search and selection procedure

uals are categorized based on their dominant traits, shown in four basic personality styles: Dominant, Influential, Steady and Conscientious. The DISC model is often used in organizational and leadership contexts to enhance communication, team dynamics, and personal development by recognizing and adapting to different behavioral styles (Meliá et al. 2024).

7 Review of existing studies

In this section, a comprehensive review of existing studies on personality detection, focusing on two main domains: firstly, present review of data collection and preprocessing, feature engineering including textual and word embeddings that directly impact model performance, demonstrating the basic steps of this methodology. Secondly, the application of ML techniques including ensemble models, and lastly, the exploration of utilizing DL algorithms including state-of-the-art transformer-based models for the detection of personality traits, as taxonomy of models' study shown in Fig. 3.

7.1 Pre-processing of data for model application

During data acquisition, preprocessing is a critical step in personality detection studies, as the quality and diversity of the data directly impact model performance. After data collec-

Table 4 Personality models and traits with behavior description

Trait	Models	High score focus	Low score focus
Openness	HEXACO, Big Five, MBTI	Curious Imaginative Open to new ideas Intuitive	Practical Prefer Routine Conservative in Thinking Sensing
Conscientiousness	HEXACO, Big Five, DISC, MBTI	Organized Responsible Goal-Oriented Judging	Flexible Spontaneous Less Structured Perceiving
Extroversion	HEXACO, Big Five, MBTI, PEN	Energetic Sociable Enthusiastic Extrovert	Reserved Introspective Prefers Solitude Introvert
Agreeableness	HEXACO, Big Five, MBTI	Compassionate Cooperative Trustworthy Thinking	Competitive Skeptical More Critical Feeling
Neuroticism	Big Five, PEN	Emotionally Reactive Prone to stress	Calm Emotionally Stable Resilient
Honesty-Humility	HEXACO	Sincere Modest Fair-minded	Self-Centered Mislead
Emotionality	HEXACO	Sensitive Empathetic	Logical Detached
Dominant	DISC	Confidence Assertive Takes Charge	Passive Prefer Supportive Role
Influential	DISC	Persuasive Motivating Charismatic	Direct Factual Less Persuasion
Steady	DISC	Reliable Patient Supportive	Adaptable Less Routine-Focus Prefer Changes
Psychoticism	PEN	Impulsivity Aggressiveness Less Empathetic Risk-Taking Unconventionality	Empathetic Cooperative Conventional Respect Societal Norms

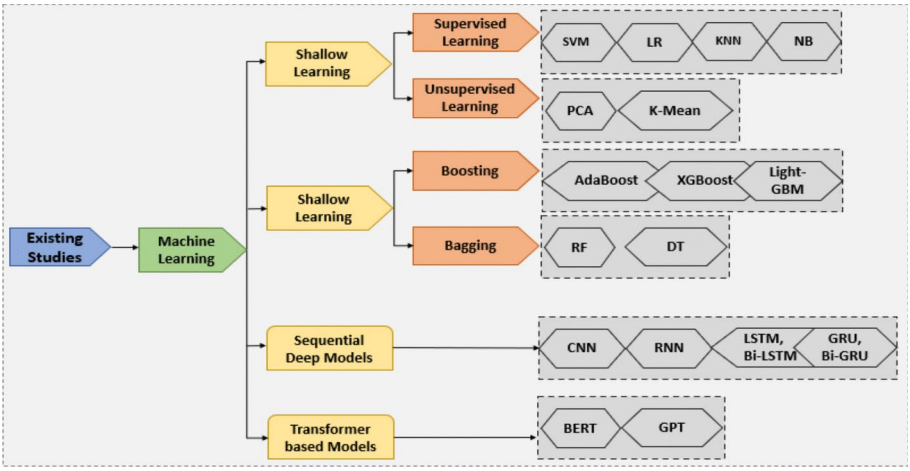

Fig. 3 Taxonomy of existing studies

tion, preprocessing steps have been taken to clean and filter the data, mentioning issues like mislaid values, and variability. The personality dataset is the base for predicting individuals' personalities through their posts and comments via extracting from social media. Before using feature extraction and selection algorithms, the dataset undertakes preprocessing. The extracted features are then trained using data mining approaches until prime accuracy is gained (Ishfaq et al. 2025).

Data preprocessing is the process of preparing raw data into a suitable format for processing, analysis, and implementation. Data preprocessing aims to improve quality of data, removal of noise, and make the data more understandable and reusable for other tasks, may involve techniques to fix noise or misleading values in dataset like often unfairly, filtering or removing borders. To clean text data from URLs, emojis, emoticons and special symbols from data (Wang et al. 2021). Splitting text into single units as tokens or words for further analysis, could be done using whitespace, punctuation, or more sophisticated methods. Once the text is tokenized, stop words, which are common words that typically do not carry significant meaning for analysis (e.g., "the", "is", "and"), are removed from the tokenized text that helps to reduce noise and focus on the more meaningful words in the text (Palomino and Aider 2022).

Converting text to a uniform case (lowercase), and standardizing variations of words to their base or root form to ensure consistency in representation involving techniques like stemming or lemmatization. The process involves removing suffixes or prefixes from words to obtain their root forms, which may not always be lexicographically correct but can be useful for tasks like search or text analysis where variations of words should be considered equivalent. Lemmatization to reduce words to their base or dictionary form, known as the lemma. Unlike stemming, which applies heuristic rules to truncate words to their base forms, lemmatization uses vocabulary and morphological analysis to obtain the correct base form of words, making it easier to analyze and interpret text data, especially in tasks related to sentiment analysis, personality assessment, and behavioral profiling (Nesca et al. 2022).

7.2 Feature engineering

Personality trait detection using classification techniques further explored using feature engineering techniques to extract and analyze meaningful data from large volume of dataset by applying various models of ML and DL. This technique helps to understand the textual and word-based relationships between sentences and individual text (Ahmed et al. 2022). It is essential to identify which text features are most important to improve performance with vectorization for converting text into numeric that algorithms can understand and process. Features are divided into two parts, text features, and word embeddings.

7.2.1 Text features

The method of representing words as dense vectors in a high dimensional space from NLP, refers to as text embedding, used to capture the contextual and semantic data about text, by enabling ML models to process and predict the language words more effectively. The Bag of Words (BoW) technique takes out unique words from text data. Each word is then represented as an independent and specific feature. BoW treats words separately. Term Frequency Inverse Document Frequency (TF-IDF) represents the attribute of personality types. Term Frequency (TF) indicates the repetition of each word in a document, while Inverse Document Frequency (IDF) manifests the presence of documents containing a specific word (Alarfaj et al. 2023). An N-gram is a sequence of N contiguous items in a text or speech sample to analyze the frequency and distribution of words or characters. In N-gram models," N" corresponds to the number of items in the sequence, which can be words, or characters. POS tagging is the process of labeling each word by assigning matching part of speech such as noun, verb, adjective etc. to words in a text data corpus. This helps in understanding the grammatical structure of context of the text (Khan et al. 2019).

7.2.2 Word embeddings

Word embeddings capture semantic links between words by replacing them in a space where similar words are closer together. Embedding approach allows for the finding of semantic similarities and contextual information within words, converting raw text data into organized and representations that facilitate more understandable language. Word embeddings are generated using algorithms like Word2Vec, GloVe, and Fast Text, which examine massive amounts of text data to learn meaningful prediction of words (Sandu et al. 2024).

Word2Vec embedding approach that teaches different depiction of words. It is built on neural networks and is trained to investigate the context of words within a given frame of text. Word2Vec finds semantic links between words. GloVe is another word embedding technique. Its aim is to learn word depiction by considering global word repetition stats. It is based on word vectors by factorizing a matrix of word repetition counts. GloVe embeddings use both semantic and syntactic information. Fast Text that considers sub word information represents words as bags of character n-grams and learns embeddings for both words and sub-words. Fast-Text is particularly useful for processing out-of-vocabulary words (Naqvi et al. 2023).

7.3 Literature review using machine learning

Sentiment analysis is increasingly recognized as a cutting-edge area within AI (Babu and Kanaga 2021). The objective of the field, effective computation, is to revolutionize and apply emotional reasoning techniques for predicting emotions across different languages (Escalante et al. 2022). The goal is to develop and predict a system capable of eliciting intuitive and emotional reactions by analyzing and converting quantifiable human elements like emotions, personality characteristics thoughts, and moods into meaningful insights (Talaat et al. 2023). The text illustrates the meaningful role of ML in presenting a method for predicting personalities, thereby deepening our understanding of human behavior (Novikov et al. 2021). Specifically, it points out the superior performance of the Gradient Boosting model, underscoring its practical value in fields like recruitment (Nirmala et al. 2024), Hate-speech analysis (Canbay 2024), team building, and personalized marketing (Chen et al. 2023).

Predicting personality traits using ML involves leveraging various algorithms to analyze patterns and relationships within data illustrated in Fig. 4. Machine learning algorithms utilize data and algorithms to train the model with the help of data and perform operations without explicit instruction. Automated personality assessment in areas like cyberbullying detection, enhanced personal assistants, specialized healthcare, and career counseling have significant studies that have employed various algorithms and computational techniques for personality assessment using social media data, including, Support Vector Machines (SVM), Naïve Bayes (NB), Logistics Regression (LR), Random Forest (RF), Extreme Gradient Boosting (XGBoost) and K-Nearest Neighbor (KNN) (Bruno and Singh 2022).

Here are frequently used ML algorithms for personality trait prediction based on literature:

7.3.1 Shallow machine learning

Shallow ML models refer to traditional ML models that have a small number of layers in their architecture. These models are often used for simpler classification and regression tasks and are commonly easier to interpret and understand but more intricate with complex patterns and high-dimensional data. Table 5 shows the analysis of existing studies used for personality traits detection using shallow ML. SVM and MNB methods demonstrate the significant correlation between behavioral tendencies among Twitter data (Cahyani and Faishal 2020), expanding data sources through Facebook exploring additional personal-

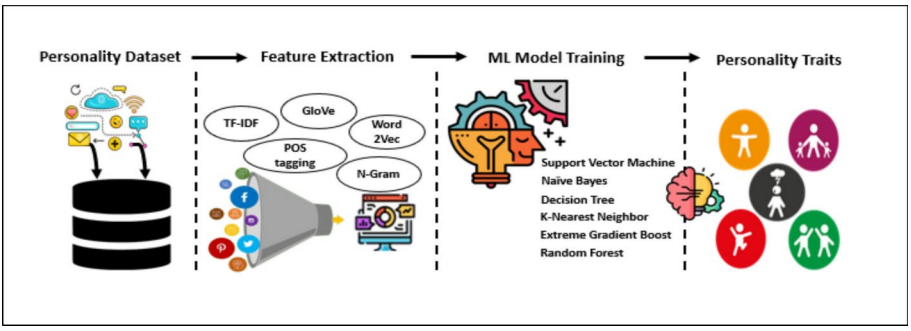

Fig. 4 Basic steps of a typical ML process

Table 5 Summary of existing studies of shallow machine learning

References	Year	Models	Features	Platform	Dataset
Carducci et al. (2018)	2018	SVM, LR, LASSO	LIWC	Facebook	My personality
Cahyani and Faishal (2020)	2020	SVM, NB	TF-IDF	Twitter	Self-created
Talasbek et al. (2020)	2020	k-means clustering	Text features	Google forms	Questionnaires
Setiawan and Wafi (2020)	2020	NB	TF-IDF	Twitter	Self-created
Annisa et al. (2020)	2020	EEG-SVM	Text features	User's brain record	ASCERTAIN
Orynbekova et al. (2021)	2021	SVM, NB,LR	TF-IDF	Twitter	MBTI
Safari and Chalechale (2022)	2022	DT, MLP, KNN	Text features	Facebook	My personality
El-Keiey et al. (2022)	2022	SVM, NB, RF, LR	Text features	Twitter	MBTI
Sharon et al. (2023)	2023	SVM, RF, LR	Text features	Google forms	Questionnaires
Kaliappan et al. (2024)	2024	SVM, DT	Text features	Blogs	Online forum

ity parameters, and integrating experts at logical insights for deeper analysis (Safari and Chalechale 2022) utilizing word embeddings and supervised learning models to analyze tweets and compute personality traits (Carducci et al. 2018). K-Mean clustering on MBTI (Talasbek et al. 2020), the NB classifier focused on the personality model based on tweet datasets (Safari and Chalechale 2022) demonstrates the new classifier for traits prediction of non-playable characters in visual novel games, utilizing the indicator for personality classification, the study develops text-based classifiers to categorize non-playable character dialogues (Pretty et al. 2024). The regression model was designed to reduce the value of personality features (William and Badholia 2021).

Using audio wave to detect the personality pattern from Electro Encephalo Gram (EEG) signals offers a new direction for research in the fields of neuroscience, psychology, and AI (Annisa et al. 2020). Another predicting personality perspective features are configured to evaluate the students' performance in academic store to detect the traits using decision tree with notable accuracy rate (El-Keiey et al. 2022). Moreover, the Apache spark platform was utilized for implementing different ML models training to detect the behavior of user for perceiving personality from tweets using word embedding techniques (Orynbekova et al. 2021) obtaining inter-dependency degrees of various personality features as trait prediction based on data dimensions in digital platforms (Choi 2021).

7.3.2 Ensemble learning

Ensemble Learning is an ML method that combines many models to enhance the overall performance and predictive accuracy of the system. The main idea at the back of the ensemble studying is to leverage the collective intelligence of more than one model instead of relying on a single model. Learning algorithms includes bagging (Random Forest, Logistics regression), stacking, and boosting (Adaboost, XGBoost). Table 6 is the analysis of existing studies used for personality traits detection.

The boosting ensemble classifier XGBoost (Amirhosseini and Kazemian 2020) is used to predict personality traits based on MBTI model using textual features such as TF-IDF on text dataset (Khan et al. 2020) providing a base for potential applications in organizational environment for recruiting and other domain services (Kamalesh and Bharathi 2022). Personality traits from social media conversations, employing supervised ML algorithms to assess traits, utilized feature extraction methods like TFIDF, GloVe, and BoW (Cherukuru et al. 2022), with TFIDF showing notable accuracy improvements on a dataset that finds that KNN, RF, and XGBoost algorithm utilized in personality prediction (Garg and Garg 2021).

Table 6 Summary of existing studies of ensemble learning

References	Year	Models	Feature	Platform	Dataset
Amirhosseini and Kazemian (2020)	2020	XGBoosting, RNN	TF-IDF	Twitter	MBTI
Mushtaq et al. (2020)	2020	XGB, SVM, LSTM	TF-IDF	Twitter	MBTI
Choi (2021)	2021	NB, RF, XGB	TF-IDF	Twitter	MBTI
Choong and Varathan (2021)	2021	LightGBM, SVM, LR	TF-IDF	Social media	café forum
Shafi et al. (2021)	2021	RF, DT, KNN	Text Features	Twitter	MBTI
Khan et al. (2020)	2022	XGB, SVM, LR, RF, MLP, KNN	N-Gram, BoW	Twitter	MBTI
Kamalesh and Bharathi (2022)	2022	XGB, SVM, LR, RF	TF-IDF, BoW	Twitter	personality
Cherukuru et al. (2022)	2022	XGBoost, LR, RF	TF-IDF	Twitter	Essays
Garg and Garg (2021)	2022	SVM, DT, KNN, LR, RF, XGB	BoW, GloVe	Twitter	MBTI, Twitter
Zhang et al. (2023)	2023	BERT+PSYLING, XGB	TF-IDF	Social media	Essays, MBTI
Sai (2023)	2023	LR, SVM, NB, RF	POS	Twitter	MBTI
Kishima et al. (2021)	2023	LightGBM	Word2Vec	Online websites	Conversation
Serrano-Guerrero et al. (2024)	2024	BERT, XGB, SVM, LR, RF, MNB	BoW, Glove, TF-IDF	Facebook	Personality

The ensemble model based on the kernel soft voting system was developed, which investigated different SVM kernels such as the radial basis function to select the most efficient point for personality recognition (Kumar et al. 2023). Moreover, judging-perceiving mapping in the dataset by utilizing character-level and word-level TF-IDF, along with features of Linguistic Exploration and Word Count (Choong and Varathan 2021). The data and classification level fusion and transfer learning from pre-trained language models were used in automated personality recognition from text integration with ensemble boosting algorithm GB to distilled version of BERT, for personality detection on Reddit (Julianda and Maharani 2023) discussing the challenges of imbalanced data and the effectiveness of various ML models, including XGBoost, SVM, and BERT (Bin Tareaf 2022).

Embeddings from Language Models, and Universal Language Model Fine-tuning in personality classification (Santos and Paraboni 2022) was evaluated on the dataset, and both annotated the big five personality traits, where bagging algorithm notably excelled in identifying traits like extroversion, conscientiousness, and openness (El-Demerdash et al. 2022). Psychological analysis (Zhang et al. 2023), the intersection of guidance and AI applying stacked ensemble techniques of deep learning DNN with boosting models (Serrano-Guerrero et al. 2024), highlighting the practical applications of ML in personal development and career planning (Fatima et al. 2022), curriculum vitae (Singh et al. 2023) analysis, highlighting the potential of various models in personality type prediction with high accuracy and efficiency of the Ensemble Bagged Trees algorithm (Shafi et al. 2021). Ensemble, unsupervised ML model utilizing models with feature selection PCA technique(Sai 2023) and others aimed at predicting personality characteristics by analyzing social media subjects in non-English languages (Alsubhi et al. 2023) with the One-Hot encoding technique focusing on the role of personality traits in guidance (Ryan et al. 2023). Advanced techniques for classification using LightGBM integrate with word embedding such as Word2Vec and Sentence BERT for feature extraction (Kishima et al. 2021). Application of ML and DL in detecting the personalities by applying k-Means Clustering with XG boosting gives better accuracy than NB (Mushtaq et al. 2020).

7.4 Literature review using deep learning

The subset of ML that handles complex patterns and structure of data with various layers of neural network to process huge data using multiple feature selection techniques required high computational training resource. Deep Models include Deep Belief Networks (DBN), Convolutional Neural Networks (CNN), Long-Short Term Memory (LSTM), Artificial Neural Networks (ANN), Gated Recurrent Unit (GRU) and Recurrent Neural Networks (RNN), and have outperformed in various domains (Mehta et al. 2020b). An analysis of how different ML and DL methods, including supervised, unsupervised, and hybrid approaches (Rahman and Halim 2022), can effectively categorize the personality traits based on text multiclass classification by considering part of compound group labels as concerned elements (Cerkez et al. 2021). For psychological analysis deep learning models such as LSTM, RNN, GRU and Bi-LSTM utilized to capture users' profiles from digital platform (Sujatha et al. 2023) and provides a future direction in text classification to detect different pattern including personality trait detection (Salminen et al. 2020). Here is a brief overview of existing studies based on sequential models and state-of-the-art transformer-based model used for personality detection task.

7.4.1 Sequential deep learning

The training of DL models using parallel processing techniques to accelerate training time and improve accuracy refers to the sequential deep learning models, enables large scale learning models on massive datasets, offering rapid experimentation and model optimization. Models include RNN, CNN, LSTM, and Bi-LSTM discussed the existing studies in Table 7. The psychopathological study was explored to detect the traits from user's generated text by applying settings of various hyperparameters including batch size, epochs, embedding sizes integrate with attention-based mechanisms of DL model Bi-LSTM to anticipate the contextual patterns for improving accuracy. The study compared the Bi-LSTM model's performance with ML classifiers like DT, SVM, KNN, MNB, XGBoost, RF, and LR, finding the most effective among them (Asghar et al. 2021). Other DL-based neural network models including bi-directional models Bi-LSTM, Bi-GRU, and sequential model such as CNN, LSTM, RNN, and GRU were carried out in literature to detect personality using You tube dataset by analyzing videos text caption (Teli and Chachoo 2023; Naik et al. 2022; Ahmad et al. 2021).

Another framework was initiated to detect patterns of personality by applying features selection to same trait group using embedded deep models (Dutta et al. 2023). An innovative method to detect learners' personalities through expressions analyzed by CNN (Bahri

Table 7 Summary of existing studies of sequential deep learning

References	Year	Models	Features	Platform	Dataset
Jeremy et al. (2021)	2020	AttRCNN-CNNs	Word and text embeddings	Facebook	My personality
Salminen et al. (2020)	2021	Bi-LSTM	Word embedding	Twitter and Facebook	Personality
Maheswara Rao et al. (2023)	2021	DeepLSTM	Fast fourier	Social media	ASCERTAIN
Silpa et al. (2023)	2022	RNN-LSTM	SMOTE	Social media	Forum
Asghar et al. (2021)	2023	RNN, LSTM, Bi-LSTM	GloVe, Fast Text, Word-2Vec	YouTube	Personality
Teli and Chachoo (2023)	2023	LSTM, Bi-LSTM, GRU	Bag of words	Twitter	MBTI
Ahmad et al. (2021)	2023	LSTM,GRU, RNN, DEC	Word embedding	Twitter	MBTI
Xue et al. (2018)	2023	LSTM	Word-2Vec	Twitter	MBTI
Ali and Ahmed (2023)	2023	RNN-LSTM	Word embedding	Twitter and facebook	Personality
Altaf et al. (2024)	2024	CNN-EEG	Word embedding	Online survey	ASCERTAIN

et al. 2023) combined with text analysis of tweets, can effectively predict personality traits (Jeremy et al. 2021), these models integrate with word embedding techniques such as word-2vec with LSTM models (Mohan et al. 2023) was used for personality detection from social network text postings, utilizing a hierarchical neural network with the ATTRCNN architecture and inception variant to extract deep semantic features evaluated on data my personality, regression algorithms yielding the lowest prediction error (Xue et al. 2018), also for recommendation system to predict personality features (Silpa et al. 2023) and CNN models, enhanced with random search optimization for improved accuracy (Ali and Ahmed 2023). The RNN-PRS model uses an RNN-LSTM framework to recommend professional careers by predicting personality types from social media activities, leveraging type indicators for personalized suggestions (Maheswara Rao et al. 2023). Utilizes a Deep Long Short-Term Memory network to categorize personality features from EEG signals by creating a new EEG dataset and comparing the performance of their model with an accuracy of 96.94 percent of ML classifiers, outperforming other models like ANN, LibSVM, KNN, and ensemble genetic programming techniques (Bhardwaj et al. 2021).

7.4.2 Transformer-based learning

DL architecture introduced another class called, Transformer-based models, these models power the self-attention mechanisms to capture long-range and global dependencies in data, for tasks like sentiment analysis, text generation, language translation, and understanding. Transformer based models, include BERT, GPT, Distill BERT. Below Table 8 depicts the analysis of existing studies used for personality traits detection.

For automatic detection using BERT model was also applied with limited dataset Essays and mypersonality mapped to personality model of big five founded with DL-based fusion

Table 8 Summary of existing studies of transformer-based learning

References	Year	Models	Features	Platform	Dataset
Chowanda et al. (2022)	2021	LSTM, BERT	Word embedding	Social media	MBTI, café forum
Lucky and Suhartono (2021)	2021	BERT, Indo-BERT	Word embedding	Twitter	Indonesian tweets
Elourajini and Aimeur (2022)	2022	AWS-EP, ELECTRA	Word embedding	Social media	Pandora, MBTI, MyPersonality
Sirasapalli and Malla (2023)	2023	BERT	Elmo, ULMfit	Social media	MyPersonality
Rauf et al. (2023)	2023	BERT	Word-2Vec, BoW, TF-IDF	Twitter	MBTI
Halimawan et al. (2022)	2023	BERT	ULMfit	Twitter	Indonesian tweets
Akber et al. (2023)	2023	BERT, SBERT	Word embedding	Twitter	MBTI
Canbay (2024)	2024	BERT	Word embedding	Social media	MBTI, big five

technique (Sirasapalli and Malla 2023) anticipate text data for extracting emotional expressions into the personality model of MBTI on Twitter data (Stachl et al. 2020), comparing the results with and without emotional cues in data applying preprocessing stage (Al-Fallooji and Al-Azawei 2022). AWS-EP and ELECTRA learning models used as encoder that allows multi-task to predict traits from text posts by setting the weights between two traits of personality models correlation of MBTI and Big five (Elourajini and Aimeur 2022), handled scattered data of MBTI personality based on sixteen dimensions with four axes mappings using Transformer-MD model (Yang et al. 2021).

Computational methods for personality prediction provide the effectiveness of combinations of different DL and ML algorithms in processing and analyzing textual data to deduce personality types by integrating the agile methodology of the Cross Industry Standardization Process for Data Mining will be used to lead the learning process (Ontoum and Chan 2022). The proposed model effectively combines emotion and personality trait detection tasks, utilizing A model-independent meta-learning framework for efficiency. The learning process significantly improves the approach, leading to root performance in both emotion and personality detection tasks (Li et al. 2022). Several computational models in NLP integrate with ML and DL approaches combining selected features for improving the text patterns (Sahono et al. 2020) using encoder BERT to predict sixteen personality axes by using online social media collected dataset based on MBTI (Rauf et al. 2023). Another model proposed with BERT with pre-trained ULMfit model utilizing feature of sentiment, augmentation, part of speech tagging for grammatical representation for data scaling performance (Halimawan et al. 2022) (Cerkez and Vareskic 2021) and aspect analyses for richer text interpretation, enhancing accuracy in personality classification from text, that shows its effectiveness through solid baseline models, hyperparameter optimization (Chowanda et al. 2022), and an emphasis on data quality and representativeness of text (Basto 2021) to anticipate personality from question–answer session using SBERT model (Akber et al. 2023). Moreover, XGBoost classifier is used with BERT transformer model to improve the performance and reliability of prediction models, particularly analyzing user behavior and traits based on social media activities (Lucky and Suhartono 2021). A fuzzy logic-based approach utilized to detect patterns of personality by integrating algorithmic logic with computation thinking of users' text for the analysis of MBTI personality recognition to intricate the complexities of personality, give a broader overview of performance as compared to traditional models (Wang et al. 2024).

7.5 Comparison of existing studies based on MBTI dataset

A detailed comparison of various models and approaches applied to the MBTI datasets, focusing on key performance metrics such as accuracy, precision, recall and f1-score, as shown in Table 9. The comparison spans across several approaches including shallow ML, ensemble learning, Sequential DL and state-of-the-art transformer-based models, providing a broad overview of model performance on dataset.

In shallow ML model like SVM, NB, LR, are evaluated, showing consistent results with accuracies ranging from 79 to 85%. The precision, recall, and f1-score indicate that these models achieve balanced performance, although the combination of NB and LR as in El-Keiey et al. (2022) achieves the highest accuracy at 90%. Moving to the ensemble learning approach, methods that combine multiple classifiers, such as XGBoosting, RF, GB are

Table 9 Comparison of existing studies based on MBTI dataset

MBTI dataset results (%)						
Sr. No	References	Models	Accuracy	Precision	Recall	F1-score
Shallow machine learning approach						
1	Orynbekova et al. (2021)	SVM, NB,LR	85	82	84	82
2	Pretty et al. (2024)	SVM, NB, RF, LR	71	70	71	71
3	El-Keiey et al. (2022)	NB, LR	90	89	90	89
Ensemble learning approach						
4	Amirhosseini and Kazemian ((2020)	XGBoosting, RNN	86	85	81	80
5	Choi (2021)	NB, RF, XGB	88	86	88	87
6	Shafi et al. (2021)	RF, DT, KNN	85	89	83	81
7	Khan et al. (2020)	XGB, LR, RF,	89	85	82	84
8	Garg and Garg (2021)	SVM, LR, RF, XGB	90	92	94	94
Sequential deep learning approach						
9	Teli and Chachoo (2023)	LSTM, Bi-LSTM, GRU, CNN	85	78	85	87
10	Ahmad et al. (2021)	LSTM + CNN, RNN	88	88	88	88
11	Xue et al. (2018)	LSTM + DEC	97	92	90	95
Transformer-based models						
12	Elourajini and Aimeur (2022)	AWS-EP, ELECTRA	78	79	74	86
13	Rauf et al. (2023)	BERT	78	75	78	78
14	Akber et al. (2023)	BERT, SBERT	90	88	89	90
15	Canbay (2024)	BERT	66	67	65	65

shown to improve results further. The accuracies in this category are notable high, with some models achieving up to 95% accuracy, precision and recall (Garg and Garg 2021). This suggests that ensemble techniques are highly effective in enhancing performance compared to individual models in shallow ML approaches. The Sequential DL shows models like LSTM, Bi-LSTM, CNN and GRU being applied to MBTI dataset. These models exhibit robust performance, achieving accuracies between 86 and 88%, with high precision and recall as well. This suggests that deep learning models capture time-dependence and relationships in the dataset effectively. Finally, Transformer-based models display varied performance using model AWS-EP and ELECTRA achieve relatively high accuracies of 78% and 75%, BERT and SBERT showed mixed results with accuracy as high 89%, but some configuration show lower performance metrics as (Elourajini and Aimeur 2022). This highlights the variability in transformer-based models depending on the specific architecture and implementation. This analysis exhibits that in comparison of all model on same dataset, ensemble and deep learning approaches tend to outperform shallow machine learning models in terms of accuracy and general performance across the various evaluation metrics. Transformer-based models, though powerful, exhibit some variability depending on how they are applied.

7.5.1 Strength and limitations of models for personality trait detection

The application of ML, and DL based models in personality trait detection presents unique strengths and limitations. Traditional ML models, such as Support Vector Machines (SVM), Naïve Bayes (NB), and Logistic Regression (LR), are widely used for text classification tasks, including personality trait prediction (Safari and Chalechale 2023). Their primary strength lies in their interpretability and efficiency on small- to medium-sized datasets. However, their reliance on feature extraction and limited capacity for capturing deep semantic relationships in text data pose challenges when analyzing complex, context-dependent personality traits derived from user-generated content (Naithani and Raiwani 2023). Ensemble learning models, such as Random Forest and Gradient Boosting, improve classification performance by combining multiple weak learners. They are particularly useful for personality trait detection as they reduce overfitting and enhance generalization across diverse linguistic styles in social media data. However, they require hyperparameters tuning and may not be as strong when dealing with highly contextual, long-range dependencies in textual data (Sivakumar et al. 2023). Deep learning approaches, including Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs), offer superior feature learning capabilities, making them highly powerful for extracting latent hidden patterns in social media text. Sequential models like LSTM networks and GRUs can intricate long-range dependencies, which are crucial when predicting personality-related text spanning multiple posts or conversational texts. However, these models require huge datasets and high computational resources, limiting their accessibility for real-time applications (Kelvin and Utomo May 2024). Transformer-based models, such as BERT and GPT, have transformed personality trait detection by integrating with built-in contextual embeddings. These models can learn complex language patterns, including false or sarcasm-based text, indirect expressions, and implicit meanings that are critical aspects of personality inference. Additionally, their ability to process long-form textual data as whole sentence from multiple sources for enhancing classification accuracy. Nonetheless, their interpretability remains a challenge, as their predictions often function as black boxes, making it difficult to explain decision-making processes. Moreover, fine-tuning transformer models require substantial labeled data and computational power (Jain et al. 2024a).

Although these models have strengths and weaknesses, the choice depends on the complexity of personality traits to be analyzed, the nature of the dataset, and computational constraints. A hybrid approach coupling with ML for interpretability, deep learning for feature extraction, and transformers for contextual understanding may offer the best balance in advancing personality trait detection research.

8 Performance evaluation measures

Personality traits prediction is measured using a variety of indicators, including precision, accuracy, and F1 score as discussed for the efficacy of datasets, and prediction models, is evaluated using these parameters. In evaluating a model's performance on dataset, key metrics are applied including accuracy, precision, recall, f1-score, ROC, and AUC (Ahmad et al. 2023). In NLP, selecting the appropriate performance metrics is essential for evaluating

classification models effectively, as each metric provides unique insights based on dataset's characteristics (Dang and Verma 2024).

8.1 Accuracy

To measure the proportion of correctly classified instances out of total number of instances. This metric provides a straightforward indicator of model performance; however, in an imbalanced dataset, relying solely on accuracy may be misleading, can be calculated as in Eq. 1. For example, in a dataset where one class dominates, a model may achieve high accuracy simply by predicting the majority class consistently (Saeed et al. 2023). To address this, it is often used in combination with metrics like precision and recall.

$$\frac{TP + TN}{TF + FN + FP + TP} \tag{1}$$

where TP, TN, FP, FN stands for True Positive, True Negative, False Positive, and False Negative, respectively.

8.2 Precision

It quantifies the accuracy of true positive prediction out of all positive predictions made by the model. This metric is particularly valuable in scenarios where false positive carry a high cost (Akber et al. 2024), computed as in Eq. 2. Task like personality detection and precision tell us how accurately no. of instances classified as having a specific trait (e.g., introverted) possess that trait.

$$\frac{TP}{TP + FP} \tag{2}$$

8.3 Recall

Indicates the model's ability to capture all true positive instances, without missing any out of all actual positive instances in the dataset (Ahmed et al. 2024), expressed as in Eq. 3. For example, to detect "Extrovert" trait and there are 100 people who are truly extroverted, but the model correctly identifies only 70 of them as extroverted, then recall is high for extrovert trait. However, if 20 extroverted individuals are missed, that means the model has some space for improvement in capturing all true extroverted.

$$\frac{TP}{TP + FN} \tag{3}$$

8.4 F1-score

The F1-Score is a single metric that combines both precision and recall, providing a balance between the two. It is useful when need of comprehensive measure of model performance, especially when there is an uneven class distribution, can be computed as in Eq. 4. If the

model predicts 50 people as extroverted but only 30 of those predictions are correct and it successfully identifies 60 out of 100 actual extroverts (recall), the f1-score will help assess how well the model is performing overall. A high f1-score would indicate that the model is effectively balancing the trade-off between identifying true extroverts and minimizing incorrect predictions (Nakib et al. 2024).

$$\frac{2(Precision * Recall)}{Precision + Recall} \tag{4}$$

8.5 AUC-ROC

Quantifies the model's ability to distinguish between the positive and negative classes across different threshold values. It reflects the overall model performance in classifying instances correctly, calculated as in Eq. 5. In a personality detection model that predicts whether individuals are introverted or extroverted, the AUC-ROC provides an indication of how well the model can differentiate between the two personality types. An AUC value closer to 1 means the model is excellent at distinguishing between introvert and extrovert users, while a value around 0.5 suggests that the model is not much better than random guessing (Zara et al. 2024).

$$\sum_{i=1}^{n-1} \frac{(FP_i - FP_{i-1}) \,.\, (TP_i + TP_{i-1})}{2} \tag{5}$$

By choosing the right metric based on the specific context and goals of NLP task, research can gain deeper insights into their models' performance, enabling more informed decision-making. When evaluating classification models for predicting personality traits, these performance metrics are commonly used, offering a multifaceted view of the model's ability to classify personality types accurately from the dataset.

9 Experimental results

The proposed methodology outlines a systematic approach, as shown in Fig. 5 to predict personality traits from textual data using a combination of preprocessing techniques, feature engineering methods, and ML as well as DL models as the components of the proposed methodology to forecast personality traits from textual data. The section is divided into four subsections that shape the general layout of the recommended methodology; each of these sections is aimed at tackling different aspects of the process. Data preprocessing helps to make the text suitable for analysis after eliminating different uninformative features in the text. In feature engineering, there are significant meaningful patterns and representations extracted from text followed by applied models for personality traits classification. Finally, the section of the dataset description gives brief information about the source data utilized during the research. Thus, this proves a perfect approach that guarantees the right application of natural language processing concepts within the personality prediction systems.

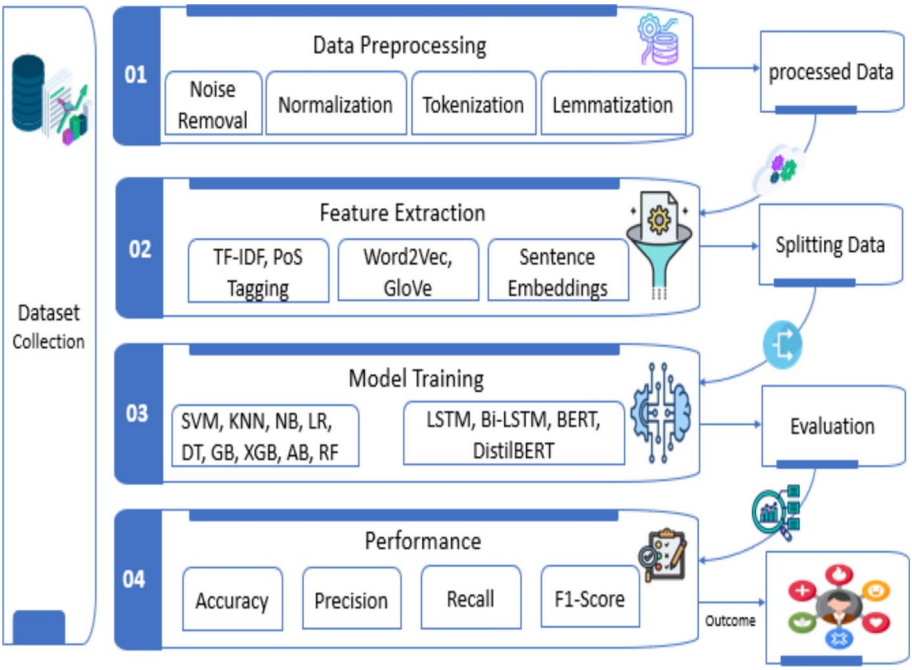

Fig. 5 Unified framework of experimental methodology

9.1 Dataset used for empirical analysis

The data used in this study sourced from online platform Kaggle named as MBTI Dataset. This corpus consists of text samples belonging to different MBTI personality types including INTJ and ENFP types. These annotations point to 16 different personalities of users that allow supervised learning techniques to predict the patterns of textual data to predict traits, contributes to the quality of the experiment, therefore, the dataset is useful for examining the correlation between text and personality trait. For prediction trait using textual dataset, data preprocessing is an important process since it cleans the raw textual data and prepares for analysis. The first step is the elimination of stop words, those are words with no particular importance toward the analysis. URLs and hyperlinks are also removed to quash noise from unrelated external links. Other symbols including pipes |||, and symbols, and all forms of punctuation marks are eliminated to reduce the text complexity. Moreover, numbers are removed from the text as often they are irrelevant to the personality traits description. Preprocessing is then performed that includes converting all text to lowercase to eliminate cases where all characters are capitalized in a sentence. In addition, lemmatization and stemming is used to convert the words to its root form to help category similar words. Last of all the method splitting the text into individual words which can be referred to as tokens enables the use of the other text analysis techniques.

9.2 Feature exploration for experimentation

Feature engineering is the process of creating useful representations of text data to feed to the machine learning and deep learning model. Two main categories of features are considered:

9.2.1 Textual features

Term Frequency Inverse Document Frequency (TF-IDF) is a technique to determine the importance of particular words used in documents regarding whole texts. This approach highlights terms that are relevant within a certain category of textual information. To determine the syntactic structure of the sentences as well syntactic features that may be linked to a personality, Part-of-Speech (POS) tagging is also used (Grunenberg et al. 2024).

9.2.2 Word embeddings

High level of text extraction is incorporated to capture semantic and contextual features from the document. The former, which is Word2Vec, assigns high dimension vectors of words depending on their contextual data while the latter, GloVe consolidates thorough worldwide coexistence statistic data to create embeddings that portray the relationship between words. Following the relative vectors, there are the sentence embeddings which aim at encoding the given sentences into vectors, to display a more general representation of the instances.

9.3 Applied models

The applied models include basic machine learning, ensemble learning, deep learning and Transformer learning models to deal with textual data for prediction. These models' purpose is to make high use of features extraction techniques and embeddings to classify the traits that given the challenge which always posed by personality prediction models.

9.3.1 Machine learning models

Several basic types of machine learning techniques currently in use are employed. SVM stands for Support Vector Machine and is a supervised learning algorithm categorized as being extremely strong in classification. Logistic regression or NB is a probabilistic based classifier which is suitable for text data based on Bayes theorem, while LR is a linear method that performs well for classification problems and DT which builds trees where data is split using feature thresholds. K-Nearest Neighbors is an instance of non-parametric learning which classifies an instance based on it similarity to close instances.

9.3.2 Ensemble models

Ensemble methods are used to improve performance rates, lift and fuse various learners. GB creates the weak learner for merging into many so that losses are minimized in an iterative manner, and XGBoost provides an enhancement for the same by optimizing for missing values and curb overfitting. RF accumulates the predictions of many decision trees so that

the final determinations are refined, whereas Adaboost different weak learners together in stages to allow the formation of a stronger model.

9.3.3 Deep learning models

Such patterns are captured using deep learning techniques to predict the data sets accurately. Text-based dependencies and long-term relations are considered using Long Short-Term Memory (LSTM) networks. This capability is extended by Bidirectional LSTM (Bi-LSTM) that treat both past and future context for each token aiding the model's comprehension of the text.

9.3.4 Transformer-based models

Transformer models are used at the current level of their higher contextual awareness. BERT for short Bidirectional Encoder Representations from Transformers is better at capturing both forward and backward relations in the text to create better brackets of contextual embeddings. Distil BERT is a cheaper version of BERT yet one that is not as computationally intensive while still ranking quite well.

9.4 Results with machine learning models

The analysis of the outcomes of the machine learning suggests that TF IDF features would be more relevant than the POS based features in all the personality aspects whatsoever, supporting the notion that the choice of features is of paramount importance to high frequency personality prediction. From the models, the results of which are shown in Table 10 revealed that SVM emerged superior, due to its efficiency at interpreting high-dimensional data, with

Table 10 Experimental results to predict personality traits using textual features (Results in acc %)

Features	Models	Extroversion	Conscientiousness	Agreeableness	Openness
TF-IDF	SVM	86	80	84	89
	NB	78	63	75	86
	LR	84	70	84	87
	DT	77	70	84	84
	KNN	79	70	70	86
	GB	86	81	82	90
	RF	79	74	79	86
	XGB	86	80	83	90
	AdaBoost	84	79	81	89
POS	SVM	76	61	63	86
	NB	70	60	58	70
	LR	78	62	62	86
	DT	64	53	54	75
	KNN	72	56	57	85
	GB	78	61	62	86
	RF	78	60	61	86
	XGB	76	56	58	85
	AdaBoost	78	60	61	78

a measured accuracy of Openness of 89% and Conscientiousness of 80%. The prior performance indicates that SVM adequately tunes the data and retains the semantic density of textual data in terms of TF-IDF vectors. Consequently, the models such as GB and XGBoost also had high accuracy bench marked across the traits, including Extroversion and Agreeableness. For instance, GB attained 86% for Extroversion and 90% for Openness, suggesting the model's reliability in managing intricate interactions inherent in the data stream.

On the hand, POS-based features provided lower accuracies as compared to text-based feature except SVM; the NB and DT provided the lowest accuracies. For instance, accuracy in only around 53% for Conscientiousness was obtained by the DT and this shows that the feature-based, syntactic patterns used lack discriminatory power for personality traits. Still, GB and XGBoost models had a rather decent performance with POS features, proving the authors' hypothesis about the ability of such models to fuse less informative features. Figure 6 and 7 shows the comparative analysis of the results obtained. In general, the results confirm that TF-IDF representations as the best way of capturing more detailed linguistic patterns that reflect the subject's personality when compared to a more basic model like Nouns, and when combined with powerful models such as SVM and ensemble models. This comparison strongly shows that when using powerful models, the TF—IDF features are significant in personality predictions as compared to the other features.

9.5 Results with deep learning algorithms

LSTM and Bi-LSTM models showed an improvement in performance by almost 25% than traditional machine learning methods. The use of Sentence Embeddings provided the highest accuracy results of all the traits with Bi-LSTM with an accuracy level of 92% in Extroversion and 91% in Agreeableness, as displayed in Table 11. Overall, better performance can be explained by the fact that deep learning models can explore hierarchical representations of textual data, thereby detecting both local and global dependencies. This feature is made even stronger by the bidirectional characteristic of Bi-LSTM because this approach allows the text sequences to be processed to the forward and backward returns, provid-

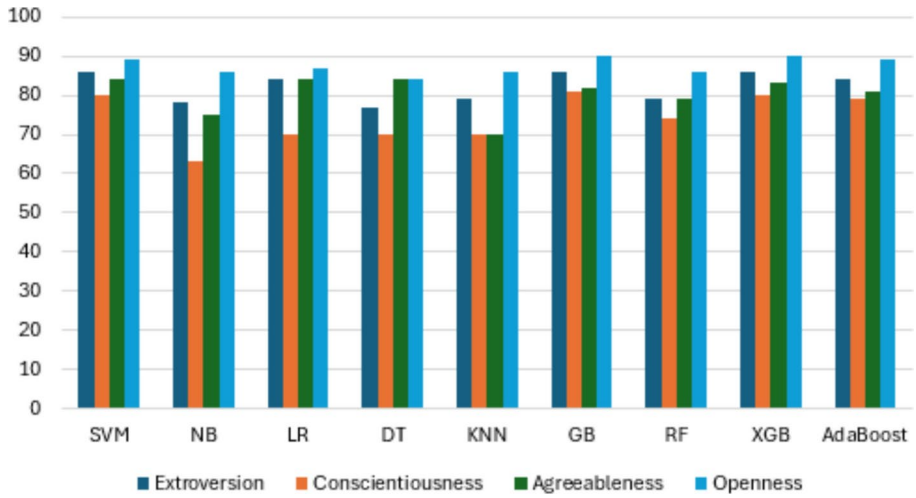

Fig. 6 Results analysis of TF-IDF feature based on accuracy measure

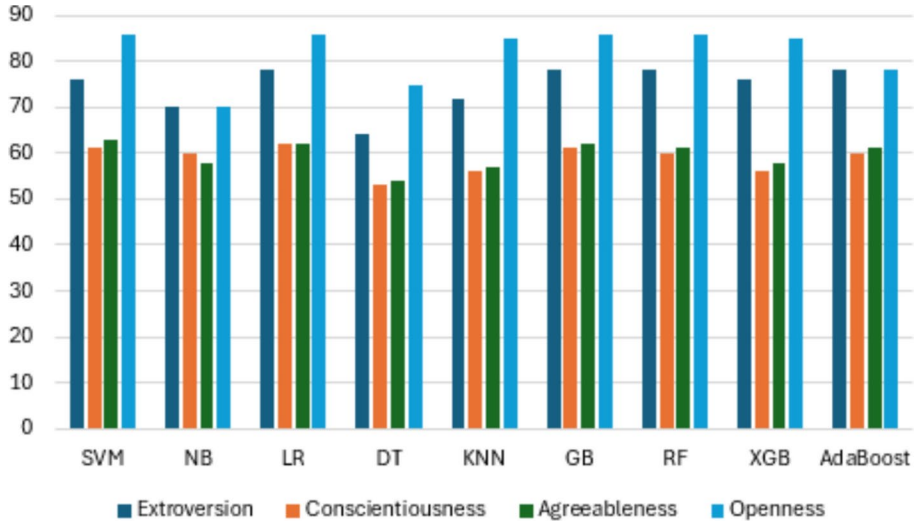

Fig. 7 Results analysis of PoS tagging feature based on accuracy measure

Table 11 Experimental results to predict personality traits using word embeddings (Results in acc %)

Model	Embeddings	Extroversion	Conscientiousness	Agreeableness	Openness
LSTM	Word2Vec	89	87	86	86
	GloVe	87	85	89	84
	Sentence Embeddings	91	89	90	88
Bi-LSTM	Word2Vec	88	86	85	85
	GloVe	89	87	91	86
	Sentence Embeddings	92	90	91	89

ing more accurate information on the contextual interdependencies. The type of embeddings used was coupled with a great influence on the performance of such models. Sentence Embeddings performed better compared to Word2Vec and GloVe for a similar reason that it can convert entire sentences with vectors and capture semantics and syntactic qualities. For instance, LSTM we used with the Sentence Embeddings yielded 91% for Agreeableness as compared to 86% for Word2Vec and 85% for GloVe. This brings out the need for a proper choice of embeddings that would match with the task at hand. It must also be noted that the results show that both LSTM and Bi-LSTM are equally good but Bi-LSTM edge is slightly better which reflects the bidirectional processing gain in personality prediction tasks.

It is clear from this analysis, from Figs. 8 and 9, to further refine deep learning models for personality prediction tasks, features such as the high-quality Sentence Embeddings are highly beneficial. Across all the embeddings, Bi-LSTM has maintained top accuracy, making it the best structure to capture any subtlety in the underlying language.

Deep learning results further attest to the role of neural architectures for personality prediction. These models manage to learn sequential dependencies and, by their nature,

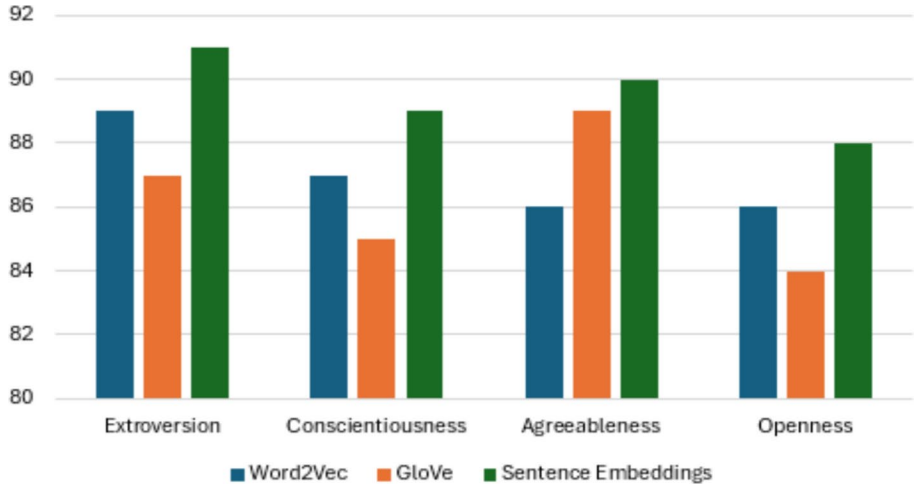

Fig. 8 Results analysis of LSTM model based on accuracy measure

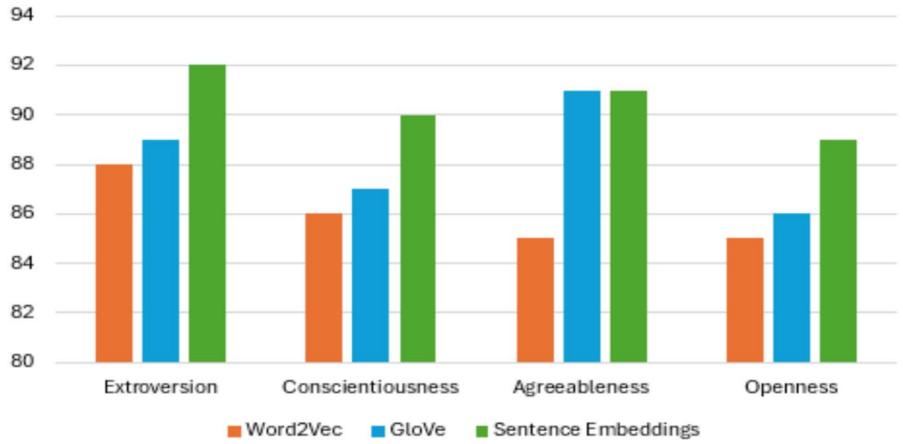

Fig. 9 Results analysis of Bi-LSTM model based on accuracy measure

improve over standard machine learning even for the traits that need more contextual information, like Conscientiousness and Openness.

9.6 Results with state-of-the-art transformer-based models

Transformer models especially Distil BERT, did incredibly well and the accuracy score set a new high for personality prediction. Distil BERT gave the highest accuracy for such parameters as Conscientiousness: 91% and Openness: 92% of overall accuracy, outperforming even more than ML and DL architectures, as defined in Table 12. This success can be attributed to the pre-trained nature of transformer models which are fine-tuned in massive corpora necessary for mastering deep semantic and contextual aspects of the textual data.

Table 12 Experimental results to predict personality traits using transformer-based model (Results in acc %)

Model	Embeddings	Extroversion	Conscientiousness	Agreeableness	Openness
BERT	Bert Encoder	82	77	82	88
Distil BERT		88	91	89	92

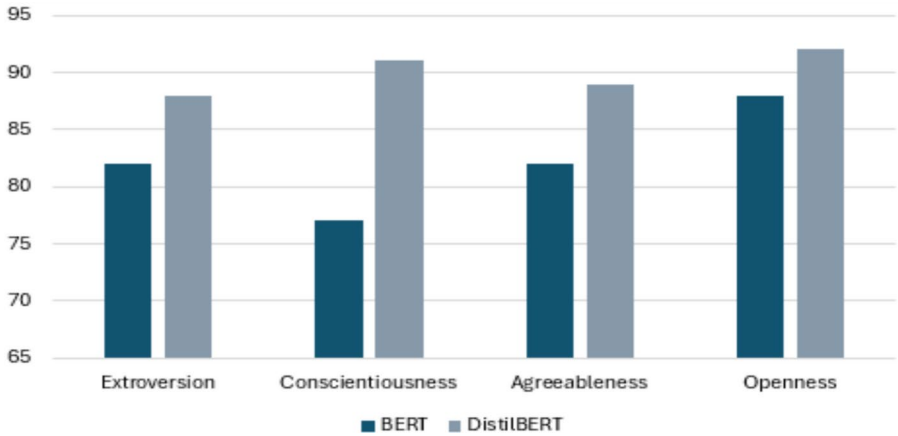

Fig. 10 Results analysis of transformer models based on accuracy measure

Transformers do not process data word by word as traditional models, and LSTMs do but then the transformer processes the whole sentence at once, which not only helps to include all the context but are also much more efficient.

Notably, there is a little difference between the results of Distil BERT and BERT when analyzing the results Distil BERT obtained 88% while BERT was at 82% in analyzing Extroversion. This result reinforces the use of Distil BERT, which has 40% of BERT model's predictive accuracy while having less computational requirements. The fact that transformers so consistently outperformed the other models on all the traits analyzed shows that this model is very good for over different aspects of personality detection, as shown in Fig. 10. Compared to other methods, transformer models demonstrate a significant switch of states in the experiments' results. The main advantage for them, which stems from the rational use of the training process of contextual embeddings and their consequent fine-tuning for tasks, is an unmatched measure of accuracy. These models work well when the trait involves analysis where wide knowledge of the ends of the text is implied where complex and ambiguous language are used.

9.7 Discussion

In the era of modernization, digitization of human personality becomes buoyant instead of being futuristic. As the number of users increases at social platforms, personality trait prediction has emerged to be an interesting research field. Towards this end, this study uses state-of-the-art approaches including shallow machine learning models, ensemble models, sequential deep learning models, as well as transformer-based models for predicting

the desired personality dimensions including extraversion, agreeableness, conscientiousness, and openness. The datasets selected, MBTI datasets, offer varied perspective on one's behavior, social media activity along with personality traits which serve as the primary strength of the work.

In our experiments, we have illustrated the advantages of one model over another in personality trait classification problem. The applications of machine learning with a limited number of layers include LR and SVM, which act like benchmarks that compare to improve prediction power, and ensemble models used to boost the prediction's accuracy include XGBoost and RF. Most deep learning models for sequential data including the LSTMs can capture temporal structure within the data. The advanced level progressive model, the transformer-based model achieved top results with Distil BERT model, accuracy of 92%, measured using standard performance evaluation such as accuracy, precision, recall, and f1-score. It is demonstrated that the ability to learn depends on the pattern of personality trait detection across different models, and this pattern changes the predictive power of each approach. The experimental results suggest that deep learning architectures (LSTM) and transformer-based models (DistilBERT) outperform others, as mostly especially using contextual and sentence embeddings. This shows that transformer models can indeed capture deep semantic meanings and contextual dependencies, which are well suited for more intricate patterns of personality inference. Similarly, LSTM yields 91% accuracy in predicting Extrovert from textual patterns which emphasizes the role of sequential information processing in modeling any personality related textual patterns. However, the predicate of this preference is the broad domain of traditional machine learning models such as RF and XGB, and their ability to utilize these structured text representations (TF-IDF) that are effective yet not ideal for uncovering more semantic relationships. That is, gradient boosting techniques improve feature selection and model robustness in structured representations, and moreover, the higher performance of XGBoost (90%) over Random Forest (86%) proves the ability of Openness prediction. These patterns imply that the prediction of a personality trait can be more effectively achieved by an intermediate balance of deep contextual understanding (i.e. transformers and LSTMs) and efficient feature learning (i.e., RF and XGB). These findings indicate a strong outcome from the fact that while traditional models are still relevant, advanced embeddings greatly enhance predictive accuracy. Transformers possess the ability to generalize over complex linguistic patterns, which provides a scalable solution for personality recognition applications, and fit well for deployment in real world application domain like HR analytics, targeted marketing, and mental health assessments (Benjamin 2025). The implementation of such models to actual world solutions has the potential to transform a wide array of disciplines from Psychology to automatic consumer targeting—with exponentially richer and more accurate insights into people's behavior.

10 Data sources and datasets

To work in any domain, it is imperative to have information about dataset being used in that domain. Table 13 displays some of the real – world datasets utilized for various studies in related literature to capture comprehensive information about individual behavior traits, specifically in English language. Datasets often comprise demographic data such as age,

Table 13 Comprehensive analysis of personality prediction datasets

Sr. No	Dataset	Source	Attributes	No. of instances	No. of posts	No. of participants	Measure	Availability
1	My-Personality	Safari and Chalechale (2022)	User profiles, psychometric, assessment, online activities	10	10,000	300	Big Five	Private
2	Conversation	Kishima et al. (2021)	daily-life conversations	5	15,269	4723	Big Five	Private
3	Pandora	Elourajini and Aimeur (2022)	Personality and demographic labels	7	16,000	10,000	MBTI	Private
4	ASCERTAIN	Subramanian et al. (2018)	Physiological signals, implicit personality, affect recognition	6	731	58	Big Five	Private
5	Facebook AI – Persona Chat	https://www.kaggle.com/datasets/atharvjairath/personachat	Comments, Conversation Logs	6	8784	93	PEN	Public
6	NeoNyx: Personality Profiling Dataset	https://www.kaggle.com/datasets/abdelkarimbsalah/nyx-2-1	Textual description, votes	11	5000	748	MBTI, Enneagram	Public
7	Essays I	https://github.com/SenticNet/personality-detection	The content of essays	5	2468	132	Big Five	Public
8	Essays II	Request granted	Big Five scores	5	2400	250	Big Five	Private
9	Big Five Personality	https://www.kaggle.com/datasets/tunguz/big-fivepersonality-test	Text, Survey Responses	5	1,015,342	400	MBTI	Public
10	Friends Persona	https://github.com/emorynlp/personality-detection	Dialogues, character interactions, Big Five labels	5	711	2	Big Five	Public
11	MBTI	https://www.kaggle.com/datasets/datasnaek/mbtitype/data	tweet texts, and user profiles	4	8600	23	MBTI	Public

Table 13 (continued)

Sr. No	Dataset	Source	Attributes	No. of instances	No. of posts	No. of participants	Measure	Availability
12	YouTube	https://www.kaggle.com/code/lisaameliewittmann/personality-profiling-of-youtube-vloggers	Video, speech, and transcripts	10	10,000	404	Big Five	Public
13	Blended Skill Talk	https://www.kaggle.com/datasets/thedevastator/unlock-the-creative-power-of-dynamic-dialogues	Conversation	8	7000	1500	PEN	Public
14	Personae	https://zenodo.org/records/4643756	Documentary about Artificial Life	3	18,000	145	Big Five	Public

gender, occupation and a few of them based on sentence behavior where individual's scores on various personality dimensions are recorded (Bozyigit et al. 2024).

Additionally, few of them may contain behavioral or situational data, such as responses to specific scenarios or psychological test, to further contextualize the personality trait. Such datasets are fully anonymized, ensuring that no personally identifiable information is disclosed, thereby eliminating any privacy concerns. Additionally, all labels within the datasets were assigned by domain experts, specifically psychiatrists, which enhances the reliability and validity of the tagging process and minimizes potential biased in labels.

10.1 FriendsPersona

The"FriendsPersona" dataset (Chen and Choi 2016), an extension of the original Friends TV Show Dataset introduces a valuable resource for studying the interplay between dialogue content and personality traits. Comprising 711 annotated conversations from the iconic "Friends" television series, each dialogue is meticulously labeled with binary indicators for the Big Five personality attributes. The dataset is 731 KB in size and available for free download.[1]

10.2 MBTI

The MBTI personality type dataset stands as a prominent and expansive resource in the realm of personality analysis, recognized for its distinction as the largest freely available personality dataset. Specifically sourced from Twitter through the Personality Café forum, this dataset encompasses a comprehensive collection of fifty tweet texts from 8675 volunteers, each associated with their corresponding MBTI personality labels. With a total of 422,845 labeled data points in the format of (post text, MBTI type), this dataset provides a substantial foundation for research endeavors in the field of personality prediction and NLP. The dataset is 26 KB in size and is freely available for download[2].

[1] https://github.com/emorynlp/personality-detection/Accessed on 16-February-2024.

10.3 MyPersonality

The MyPersonality dataset stands as a notable and comprehensive resource in the field of personality psychology, offering valuable insights into the relationships between online behavior and individual traits that occurred in 2018. Originating from the MyPersonality Facebook application, this dataset comprises a substantial collection of user profiles, encompassing diverse demographic information, psychometric assessments, and online activities particularly focusing on the Big Five personality dimensions through online interactions, status updates, and other digital engagements with 10,000 posts from 250 Facebook users, and another 150 Facebook users to navigate alternative sources and datasets for studying the connection between online activities and personality attributes. This dataset was created by the Cambridge Psychometrics Centre. This dataset seems, however, to be currently unavailable.

10.4 Essays

The"Essays" dataset, also called essays-I (Pennebaker and King 1999), created by Pennebaker and King in 1999, is a collection of 2,468 anonymous essays, each tagged with the author's self-reported Big-Five personality traits. These stream-of-consciousness writings were written by volunteers in a controlled setting, and the authors were expressly instructed to name their personality traits using the framework. The size of the dataset is 7.8 MB and can be downloaded for free. The enhanced version of dataset Essay II dataset (Tausczik and Pennebaker 2010), comprises 2,400 essays that underwent manual labeling with personality scores for the same Big-Five traits, not only contains the textual content of the essays but also incorporates refined and classified personality information. Dataset seems, however, to be currently unavailable.

10.5 Facebook AI: persona chat

The Facebook AI dataset is a valuable resource to enhance the engagement and specificity of chit-chat models by conditioning conversations on assigned persona profiles. The dataset comprises conversations between Turkers, where each participant is given a specific personality. The dataset focuses on two aspects: Conditioning dialogues on the assigned profiles information of each participant and utilizing information about their conversation partner, even when the latter is initially unknown. Notably, the data has been sourced from the public URLs of Kaggle repository having size of 7.99 MB, where new data are openly accessible.[2]

10.6 ASCERTAIN

The ASCERTAIN dataset (Subramanian et al. 2018), aptly abbreviated as "A multimodal databaASe for impliCit pERsonaliTy and Affect recognitIoN," is a specialized collection tailored for investigating the interrelation between emotions and personality traits. Specifically designed to explore the domain of personality and affect recognition, ASCERTAIN integrates multimodal data sources, emphasizing physiological signals. The dataset encompasses contributions from 58 users, providing a diverse and comprehensive perspective on

[2] https://www.kaggle.com/datasets/atharvjairath/personachat/Accessed on 25-October-2024.

the intricate connections between physiological responses and the expression of personality traits. This dataset is currently no longer available.

10.7 Big five personality

The Big Five textual dataset is based on responses to a questionnaire administered using the internet platform open psychometric. It contains 1,015,342 tuples organized around the five-factor model or the OCEAN model for personality characteristics. The dataset is 167 MB in size and is freely available for download.[3]

10.8 NeoNyx dataset

The dataset is a comprehensive collection of personality profiles designed for exploration and analysis of traits. It includes rich textual description and classifications across various personality typing systems, sourced from reputable databases and thoroughly cleaned for accuracy. The dataset consists of main file named "combined_data_highest_voted.csv" featuring 18 columns from which 8 columns are unique identifiers, MBTI personality type and detailed profile description, along with metadata for categorization and voting. The dataset size is 10.29 MB and is freely available.[4]

10.9 YouTube

The YouTube dataset comprises personality scores assigned to 404 YouTube vloggers. To collect this data, Amazon Mechanical Turk, and the Ten-Item Personality Inventory (TIPI) were used for explaining the recordings where individuals engaged in webcam discussions on different topics extracting text from captions and posters. The dataset closes a variety of multimedia objects, including videos, speech recordings, and script converted to text. This large dataset facilitates the exploration of the relationship between personality behavior and online content creation on the YouTube platform. The dataset size is 2.68 MB and is freely available.[5]

10.10 Blended skill talk

Blended talk is a dataset comprising 7000 conversations designed to explore various dialogue models, including personality expression, empathy, and knowledge demonstration. Each record is structured with fields for personas, context, previous utterances, free messages, guided messages, and suggestions, providing a rich foundation for stimulating conversations platform, enhancing its relevance for conversational AI research and development. The dataset has three folders splitting into test, train and validate.csv file with 16.57 MB in size and is freely available for download.[6]

[3] https://www.kaggle.com/datasets/tunguz/big-five-personality-test/Accessed on 16-February-2024.

[4] https://www.kaggle.com/datasets/abdelkarimbsalah/nyx-2-1/Accessed on 25-October-2024.

[5] https://www.kaggle.com/code/lisaameliewittmann/personality-profiling-of-youtube-vloggers /Accessed on 16-February-2024.

[6] https://www.kaggle.com/datasets/thedevastator/unlock-the-creative-power-of-dynamic-dialogues/Accessed on 25-October-2024.

10.11 Personae

The Personae dataset is a collection of essays authored by 145 undergraduate students, each approximately 1400 words in length. The essays hub on a documentary centered around the topic of Artificial Life. This collection provides worthy stats into how ones at the bachelor's level articulate their ideas and perspectives on the intersection of artificial life and documentary filmmaking. The dataset's content offers a rich source for exploring diverse viewpoints and expressions relevant to this context. The dataset is 2.3 MB in size and is freely available for download.[7]

10.12 Conversation dataset

The Conversation Dataset consists of junk extracted from daily life meetings obtained from ninety-six individuals' attire with an Electronically Activated Recorder (EAR). In this dataset, there are a total of 118,259 words and 15,269 conversation utterances, providing a lavish and diverse collection of linguistic data seized in natural, real-life settings. This dataset serves as a beneficial resource for studying and understanding patterns, dynamics, and characteristics of spoken language in genuine daily-life interactions. This dataset is no longer available now.

10.13 Pandora

The Pandora dataset consists of more than seventeen million comments, along with personality and demographic tags, encompassing data from over 10,000 Reddit users. Remarkably, the dataset includes Big Five personality trait labels for 16,000 users, making it the dedicatory Reddit dataset to incorporate such large-scale personality information. Additionally, the dataset encompasses other personality models, including the MBTI framework. On request available at.[8]

11 Applications

The enriching growth of the internet and digital media has significantly heightened the importance of detecting personality traits from text. This development is gaining attention from technologists, business leaders, government bodies, and political analysts who see its potential to enhance decision-making processes in various sectors, including business and public relations. The proliferation of social media platforms has provided a wealth of data for analyzing online social behavior either positive or negative (Ahmad et al. 2023), encouraging researchers to delve deeper into understanding users' personalities based on their digital footprints. Personality trait detection from textual data significantly impacts various fields by opening and suggesting new research directions and applications with the help of machine and deep learning model predictions. Personality trait recognition has garnered

[7] https://zenodo.org/records/4643756/Accessed on 16-February-2024.

[8] https://psy.takelab.fer.hr/datasets/all/pandora//Accessed on 16-February-2024.

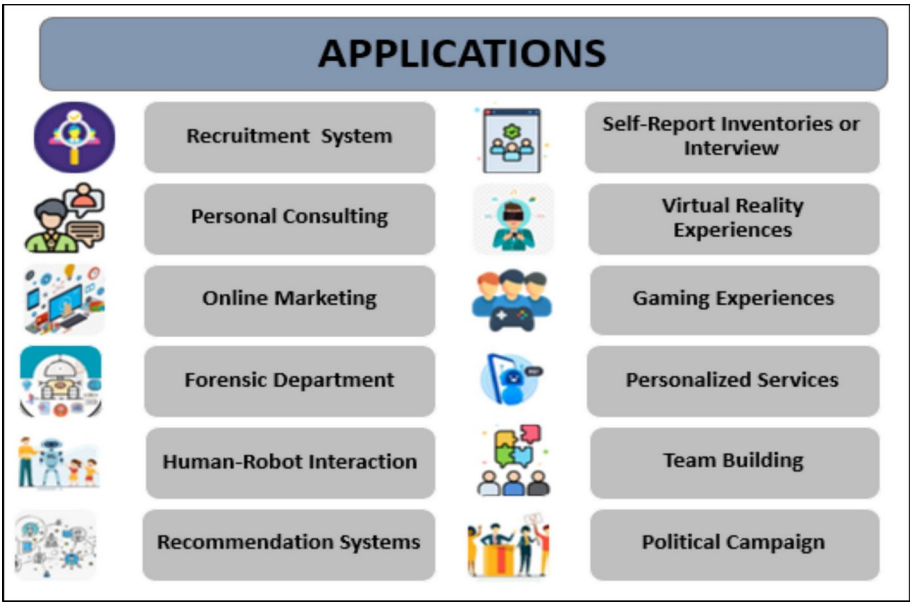

Fig. 11 Applications of personality traits detection in different domains

significant attention across diverse influencing various domains with its potential impact applications as shown in Fig. 11.

11.1 Machine learning-based approaches

Personality recognition using ML techniques is applied in various areas such as recruitment systems, career counseling, team building where ML models help to predict the best fit choice based on personality traits using traditional models such as SVM, KNN, NB, LR, and DT. These techniques widely used due to their simplicity, interpretability, and effectiveness in smaller datasets, which helps in understanding personality for improving users' interaction in digital environment. Below Few applications are employed to detect personality with the help of machine learning models.

11.1.1 Personality traits in recruitment and marketing applications

In recruitment systems, understanding candidates' personality traits aids in predicting job performance and cultural fit regarding business terms for customers also (Khan et al. 2020). Online marketing campaigns become more engaging and effective when personalized to consumers' personality traits, ensuring targeted advertisements resonate with specific audiences (Yang et al. 2023). The analysis of recommendation systems is enhanced by incorporating personality traits, leading to more accurate and satisfying content suggestions. Deception detection relies on personality cues, enabling the identification of inconsistencies in communication styles (Bucur et al. 2024). Authorship attribution sentiment analysis and opinion mining based on their research interest give benefit from insights into writers' personalities, providing a deeper understanding of content distinctions (Urooj et al. 2023).

Self-report inventories and interviews become more accurate and insightful with personality trait recognition, aiding in psychological assessments (Asghar et al. 2021). Advertising agencies tailor campaigns to specific audiences based on personality traits, increasing resonance and impact. Political campaign strategies benefit from understanding voters' personalities, allowing campaigns to tailor messages, advertisements, and outreach efforts for a more personal connection and influence (Stachl et al. 2020). However, Ethics and risk issues occur upon integrating AI-driven personality prediction in industrial applications like in HR system and marketing technologies as well as in social media analysis. In HR systems, there is a risk of biased hiring decisions if models inadvertently favor certain personality traits, leading to discrimination (Lee et al. 2023). In marketing, AI in personality profiling may lead to hyper personalization of ads, credit card frauds and the changing of consumer behavior issues in this regard regarding privacy and consent (Yang et al. 2022). Misusing personality insights in social media analysis can help to spread targeted misinformation or digital surveillance which may attack or violate user autonomy (Pillow et al. 2025). To mitigate these risks, it is crucial to empower bias mitigation approaches for maintaining fairness, accountability and regulatory compliance as well as transparent ethical AI models deployment.

11.1.2 Personality-based educational and counseling systems

Personal consulting systems benefit from tailored advice based on individual traits, enhancing the effectiveness of guidance. Career counseling benefits from personalized advice tailored to individuals' traits, guiding informed career decisions (Ahmed et al. 2023). Team building efforts are optimized by assembling diverse teams with a mix of complementary personality traits (Choong and Varathan 2021). In conflict resolution and negotiation, understanding the personality traits of involved parties' aids mediators in tailoring effective interventions. In education, adaptive learning platforms leverage personality traits to tailor teaching styles, content delivery, and assessments, optimizing the educational process for effective learning outcomes (Nisha et al. 2022).

11.2 Deep learning-based approaches

Deep learning approaches have gained popularity in personality recognition because of their ability to automatically learn complex patterns from large dataset without need for manual feature extraction. DL models excel in more dynamic and interactives applications like chatbots, virtual reality, and gaming where they adapt to users' personality trait to enhance engagement and personalization. Below applications are employed to detect personality using deep learning approaches.

11.2.1 Personality detection in health and lifestyle applications

Personal development strategies become more effective when aligned with individual traits, fostering meaningful self-improvement (Singh and Singh 2024). In forensic departments, architectural designs (Schönfelder et al. 2024), personality recognition aids in criminal profiling and understanding motives (Igwe and Durrheim 2024). Mental health diagnosis benefits from insights into patients' unique psychological needs, informing personalized

treatment plans (Yang et al. 2021). Wellness and lifestyle apps offer personalized health recommendations by considering users' traits and suggesting fitness routines, dietary plans, and stress management techniques aligned with individual preferences (El-Demerdash et al. 2022).

11.2.2 Personality recognition in interactive and visual system

In human–robot interaction, recognizing user personality enhances the design of robots, creating more intuitive and responsive interactions. Personalized services, such as chatbots (Naqvi et al. 2023), provide more engaging interactions in videos instead of traditional-based interviews (Koutsoumpis et al. 2024) to recognize user personality traits and create supportive user experience (Sharma et al. 2021). Virtual reality experiences become more immersive with personality trait recognition, as VR applications adapt scenarios and interactions based on users' traits. Social influence and trend analysis utilize personality recognition to identify influential individuals and predict trends and ideas within social networks for marketers and sociologists (Kumar et al. 2023). In gaming, personality trait recognition enhances user experiences by personalizing game-play dynamics, challenges, and narratives (Shah et al. 2021). Smart home automation leverages personality traits to adjust lighting, temperature, and entertainment systems (Bahri et al. 2023). Language learning platforms adapt teaching methods, content, and cultural insights based on learners' personality traits (Liu et al. 2024), enhancing language acquisition, and making learning more enjoyable (Basto 2021).

These applications underscore the versatility of personality trait recognition, highlighting its potential to transform and customize experiences across education, politics, psychology, human–computer interaction, wellness, entertainment, and technology in various fields. As research progresses, the integration of personality insights into these applications continues to evolve, offering new dimensions for customization and personalization.

12 Open challenges and research issues

The domain of personality trait prediction presents lots of challenges and issues that point out the limitations of existing studies in the relevant literature. In this section, we present the most updated challenges.

12.1 Utilizing LLM-based models for enhanced predictions

Leveraging LLM including BERT, GPT-3, GPT-4, PaLM and other proprietary and open source LLMs, presents opportunities for a more enhanced analysis of personality traits prediction by fine-tuning of these models. However, challenges include ensuring domain relevance and adapting the model to the intricacies of professional language, by managing their large computational requirements, and addressing issues of interpretability and ethical issues (El-Keiey et al. 2022).

12.2 Deeper analysis of word usage and intended meaning

Analyzing word usage requires a deeper understanding of context and meaning. Assigning weight to word changes and considering the influence of tone, stress points, and enunciation on the intended meaning are all challenging tasks, especially in sensitive subjects such as mental health. Expanding the analysis to incorporate tone of voice, captured through recorded speech data, introduces challenges in accurately interpreting spoken nuances. Distinguishing between variations in word pronunciation that convey different emotions poses complexities for mood analysis (Bharadwaj et al. 2018).

12.3 Utilizing deep semantic features for improved accuracy

Integrating deep semantic features into regression algorithms offers potential improvements in accuracy. They include not actual, frequently occurring words but rather deeper, non-taxonomic features that are associated with micro and macro aspects of individuals' personality traits, including tone of voice and implied meanings. The risk is to create deep semantic models to grasp the intention, feeling, and reference that often necessitate more sophisticated practices like latent representations in transformers such as BERT or GPT, also regression models that successfully use these variables without overfitting or compromising generalizability across many people (Xue et al. 2018). Secondly, human communication differs significantly between people, space and time, thus few models of personality description can fully encompass variability of semantic meanings. These limitations call for the development of more complex and contextualized algorithms, to better match the complexity of the task, as well as the collaboration of experts from natural language processing, social psychology, and computational cognitive science. (Kusumaningrum et al. 2024).

12.4 Incorporating Facebook reactions for enhanced predictions

Using semantics from Facebook responses shows issues in preserving privacy while gaining information. Furthermore, including aspects like views, emotions, readability, and formality entail advanced modeling techniques (Safari and Chalechale 2022). These reactions are distinct emotions that do not necessarily equal personality characteristics, and depend on content genre, culture, and user activities (Zhou et al. 2022).

12.5 Multilingual analysis for personality categorization

Enlarging personality prediction to languages such as Urdu and Pashto presents difficulties due to data availability, linguistic complexities, and cultural differences, and how these variables affect personality prediction across languages. Majorly, English based language datasets are used to predict as high resource, but low resource data is still unavailable. Creating strong models that can be generic across several languages and cultural conditions is a significant task (Cherukuru et al. 2022; Jain et al. 2024b).

12.6 Personalization vs. generalization

One of the biggest barriers to accurate personality identification is the balance between personalization (capturing individual-specific traits) and generalization (applying models across diverse populations). The fact that Models tailored for individual users may fail in broader contexts while generalized models' risk lacking complex and unique individual' characteristics (Liao et al. 2024).

12.7 Dataset availability for advances personality analysis

The absence of publicly available datasets and the lack of significant annotated datasets pose a significant challenge in advancing research on personality. Without access to numerous and properly annotated datasets, researchers face limitations in training and validating ML and DL models and algorithms for evaluation. The shortage of such datasets hinders the development of strong and generalizable strategies for investigating non-textual modalities, and physiological signals (Ilmini and Fernando 2024).

12.8 Lack of attention models in personality detection

Attention mechanisms, known for improving both model accuracy and interpretability by analyzing the linguistic patterns, have not yet been fully explored in this domain. Although, attention mechanisms have demonstrated significant in NLP tasks by dynamically focusing on the most relevant parts of input, their application to personality detection often reflected in subtle context-dependent cues within text (Christensen et al. 2020; Saviola et al. 2020).

12.9 Ethical and privacy consideration in AI-based personality recognition

AI-Based Personality detection model face significant ethical challenges, particularly in terms of bias, privacy, and security. Computational models often rely on sensitive personal data, such as social media activities, or behavioral patterns, raising privacy concerns about unauthorized access or misuse of personal data. The collection, storage, and processing of such data comply with privacy protection laws as GDPR, ensuring user consent and data anonymity (Islam et al. 2024). Additionally, security vulnerabilities also expose personal data to breaches, making encryption and secure storage essential which is still under-explored in existing studies. Techniques such as data anonymization, secure multi-party computation, and privacy preserving ML are crucial for protecting user data. Furthermore, addressing bias, fairness constraints and bias removal is still a gap which can be explored using explainable AI to enhance the transparency and accountability in the deployment of AI-based models to detect the patterns of personality from digital environment (Jenifa et al. 2024; Manroop et al. 2024).

For the bias mitigation various techniques designed for reducing unfair or discriminatory outcomes in AI models, so that the outcomes are fair and unbiased. Such as *Adversarial Debiasing* is the process by training a personality prediction model as well as an adversarial network which actively minimizes bias while preserving relevant personality traits. The analysis of hypothetical scenarios forms the basis on which fairness based on *Counterfactuals* ensures that personality assessments model predictions do not change considering

changes of sensitive attributes like gender, socioeconomic background, preventing biased profiling or ethnicity (Jiang et al. 2024). Additionally, *reweighting* is another method, which reweights training data to ensure representation (Waris et al. 2024) while *fair representation learning* is one which transforms input data to a more neutral state before processing (Shah et al. 2019). Implementing these techniques in personality recognition enhances fairness, transparency, and trust, ensuring ethical AI applications in academic, professional, and psychological assessments.

12.10 Addressing ambiguity in personality labels

Personality has an inherently subjective characteristic and shares numerous attributes with other personality types making it complex in classification. For instance, extraversion and agreeableness are not only related but also their lexical correlations may also be rather like each other allowing models to distinguish between them insufficiently. More specific instructions that define what kind of annotation should be provided are required together with context-dependent models (Alsini et al. 2024).

12.11 Integrating text with behavioral information

Firstly, personality is shaped not only by language but also the overall interaction, including online activity, response time, or interaction activity rates. Combining textual information with behavioral data might enhance the predictions but raises issues and various concerns about feature extraction, data gathering, and privacy preservation (Fyffe et al. 2024).

12.12 Social engineering attack implication on personality trait detection

Social engineered attack that manipulates the personality of a person using a particular trick by leveraging psychological insights to deceive individuals. Attackers provide input to exploit personality estimations or even take advantage of weaknesses inherent in the acquisition of data sets. To address this challenge strong countermeasures must be taken to make personality trait detection models less vulnerable to adversarial manipulations as well as to keep unethical practices in data processing and handling (Huseynov and Kose 2024).

Security can be enhanced by proposing attack mitigation strategies including *adversarial training* can be used in personality recognition systems to defend against social engineering attacks to enhance model robustness by exposing it to deceptive inputs and *detection mechanisms*—such as anomaly detection or behavioral analysis—with which malicious activities in the real time can be identified (Naz et al. 2024b). Additionally, *differential privacy* keeps sensitive user data safe by introducing controlled noise while the *access control mechanisms* limit the unauthorized data manipulation to ensure that the use of AI in personality assessments remains secure and ethical (Miller et al. 2020).

12.13 Incorporating multimodal data fusion

Exploring methods to integrate various data sources to predict personality features by analyzing multi source heterogenous data like personality images, situational cues, and videos data that can translated into text (Chaurasia et al. 2024). NLP involve text analysis, multi-

modal data extends NLP focus by extracting text from other data modalities such as video captions, speech to text transcriptions and text embedding in images (Agastya et al. 2019; Mahandule et al. 2023). The inclusion of multimodal data fusion provides improved possibilities for integrating different types of data, including image, video, and text, in problem-solving situations, for example, in a classification problem of predicting personality traits. To do this, a Vision Transformer (ViT) will be used for image and video processing since this approach lends itself well to capturing global contexts efficiently (Vásquez and Ochoa-Luna 2021). Video datasets are processed using frame sampling or spatiotemporal modeling to allow dynamic assessment of traits. It was found that images can be preprocessed, and data augmentation can be applied to increase feature variety. Multimodal fusions like concatenation or attention-based fusions can incorporate phenomenon from all these modalities to output reliable predictions about personality traits encompassing both static and dynamic characteristics (Agrawal et al. 2021). For images, the deep learning models can be CNNs or ViTs or others that extract geometric and or color-based features at a high level for the target task as emotion recognition or personality prediction. For example, an experiment could include comparing images of faces in different affective states to emotional categories then combining this database with the text sentiment classification for multimodal perception. Some format in audio data such as spectrogram representation of the data can then be fed to models like LSTM or transformer networks to capture temporal aspects in speech such as tone or pitch for stress or mood prediction. For the practical implementation, feature maps from text, image, and audio modalities are integrated by using attention mechanisms or feature concatenation to promote cross-modal integration. For instance, integrating video frame analysis with audio tone detection enhances the gradient level of accuracy in emotion recognition systems in human computer interfaces (Guo et al. 2024).

These attempts highlight the difficulties and challenges associated with advancing personality prediction techniques. Addressing these difficulties is imperative for the effective integration of advanced technologies and methodologies into the field that ensuring more accurate and culturally inclusive personality evaluations.

13 Brief answers to research questions

This section provides clear answers to the research questions outlined in Sect. 1.4, establishing a connection between the findings and the overall objectives of the study.

RQ1 *Which ML models are more suitable for personality traits detection in online textual content?*

Answer: Based on the analysis of existing studies, traditional ML models such as SVM, KNN, NB, and LR, have shown promising contribution in personality detection. Leveraging its scope to ensemble learning models including boosting, bagging and stacking models effectively enhance the linguistic patterns using textual features such as TF-IDF, PoS tagging, BoW and N-Gram to classify the personality traits with reasonable accuracy depending on the dataset and specific characteristics of the text being analyzed.

RQ2 *Which DL models are more appropriate for personality traits detection in online textual content?*

Answer: Advanced deep learning models, particularly those based on transformer architecture like BERT and GPT, have emerged as the most effective approach for detection of personality trait. These models excel in capturing complex linguistic and contextual patterns in data using word embeddings features such as word2vec, GloVe, Fast Text and advanced sentence embedding rely on word choice and sentence format, significantly improving predictive accuracy over traditional ML applications. The ability to fine-tune these models on specific personality datasets enhances their performance.

RQ3 *How do diverse features and machine learning based models impact the accuracy of personality trait prediction?*

Answer: By applying advanced deep embeddings, such as Word2Vec, GloVe, and sentence embeddings, along with top-performing models like BERT and Distil BERT, significant improvements in personality prediction accuracy can be achieved. These embeddings effectively capture semantic and contextual nuances in textual data, allowing models to better understand and classify personality traits. Experimental results demonstrate that leveraging contextualized embeddings enhances predictive performance, particularly for the openness trait, which exhibits high accuracy due to its strong linguistic associations in text. This highlights the crucial role of deep embeddings in refining personality recognition models, enabling a more precise and reliable assessment of individual personality traits from textual data.

RQ4 *What are the features and source of available dataset on personality trait prediction?*

Answer: Key features utilized for personality trait prediction include linguistic indicators such as sentiment, complexity, and thematic content, derived from text. Available dataset provides diverse samples along corresponding personality trait labels, that enable robust training and evaluation of both ML and DL models.

RQ5 *What are the most potential application areas for AI-based personality trait prediction and how do they impact various fields?*

Answer: AI-based personality trait prediction has significant applications across various filed contribute insights that aid in understanding of users' personality in digital environment. The integration of insights from diverse areas enhances the accuracy and depth of trait by analyzing patterns in language, emotional expressions and contextual nuances for individual differences. Moreover, drawing on various methodologies and theoretical framework allows for the development of more predictive models to improve the outcomes. This comprehensive approach enriches the understanding of personality and facilitates more precise modeling technique that help in different areas.

RQ6 *What are the main open research challenges and their applicable strategies to be employed to address these issues?*

Answer: Key challenges ensuring the ethical use of personality data, addressing data privacy issues, and improving model interpretability. Strategies to mitigate these challenges involve implementing transparent data handling practices, developing ethical guidelines for personality assessment, and utilizing explainable AI technologies to clarify how models derive their predictions. Future research should focus on establishing best practices for the responsible use of AI in personality predictions.

14 Conclusions and future directions

Personality traits are one of the trending domains of psychology and have the potential to integrate with the computer science field. In the survey, we have discussed a brief base knowledge of personality traits and relevant computational data along with tables to strengthen our arguments. The study has also mentioned the highest count of the latest and specific datasets with preferred sources and performance evaluation metrics to evaluate them, which will further help the researcher to get all information related to AI traits in one article. Moreover, the examination of feature engineering techniques highlights the significance of text embedding techniques in capturing the language and facilitating accurate predictions of character trends. Our contribution shows progression in the field by sharing review of recent studies also describing details of datasets and applications and highlighting key research gaps which can serve as potential research challenges. This study advances the application of NLP to improve the detection of personality trait. By addressing gaps in existing literature such as a limited focus on psychological interpretation of textual data and the use of advanced NLP models, this study contributes to the grounding and accuracy of personality detection to highlight the NLP models, how are used to capture language markers associated with different personality traits, aligning these with personality frameworks. By addressing research questions, our study highlights the critical factors influencing AI-based personality trait prediction, identifying promising applications areas, highlighting challenges, emphasizing the significance of psychology with computational models. In the end, several future directions and open challenges have been exploring for the investigators to conduct the computational model in the domain, including leveraging advanced technology like LLM, GPT-3, and addressing ethical concerns, by providing a holistic overview and actionable insights. In short, our article aims to look at the massive growth of psychology-based topics to integrate with ML and DL methods and the power of psychology existing theoretical based models to enhance the performance of prediction and detection.

The primary focus in this study is to focus on natural language content for personality trait prediction using NLP features and focusing on applications of ML and DL algorithms. In the future, we aim to carry out such extensive review analysis of existing studies of personality trait detection using image and video data as these data types need completely different set of visual features and latest algorithms and techniques of Digital Image Processing and Computer Vision. The latest advancements in digital imaging such as application of deep learning like vision transformers will make it comprehensive for multimodal data such as audio, video, and images. Furthermore, factors such as trust, transparency and reliability in AI-based personality detection model predictions are dependent on improving the interpretability of such model for enhancing interpretability. Thus, Explainable AI (XAI) techniques can also be added to ensure clarity into how models analyze user-generated con-

tent and derive personality traits, making their decision-making process more understandable to researchers and end-users, in various fields, like In psychological assessment to help counselors for understanding AI-generated insights into personalized therapy. Additionally, In HR management, explainability makes sure that hiring decisions are not biased and are fair. Social media platforms use XAI to enhance algorithmic transparency in content recommendations for preventing biased profiling (Panfilova and Turdakov 2024). In the education domain, explainable models empower personalized learning by offering interpretable explanations based on the student's behavior. XAI aids in ethical AI-based profiling and threat detection in the areas of forensics and security. To measure the understanding of the model by methods like SHAP (Shapley Additive Explanations) that helps in quantify each feature's contribution to predictions, offering a global and local understanding of model behavior. However, LIME (Local Interpretable Model Agnostic Explanations), generates interpretable approximations for individual predictions, making complex deep learning models more understandable, detect bias in model, and ensure the AI is ethically deployed (Gafarov et al. 2024). Using these techniques helps achieve fairness, reduces bias, and will also reinforce user trust to spread responsible AI adoption in academic, professional and psychological assessments.

Author contributions Anam and Hikmat have written a major part of the paper under the supervision of and Ali. Muhammad and Ali have helped design and improve the methodology and wrote the paper initial draft with Anam and Hikmat. Amal and Tariq have helped in improving the paper sections, such as, review methodology, datasets, performance evaluation and challenges and future directions. Ali, Amal and Tariq have improved the technical writing of paper. All authors are involved in revising the manuscript critically and have approved the final version of the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interest The authors declare no competing interests.

Ethical approval This is reviewing an article based on different existing studies, Ethical Approval was not required as no humans were involved in the study. There are no ethical issues or concerns regarding humans or animals.

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